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Faculty of science

Biology& Biotechnology Department

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Industrial Biotechnology

Laboratory Manual

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Industrial Biotechnology

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Introduction

Industrial Biotechnology is basically associated with the commercial exploitation of microbes for the benefit of mankind. These microbial products may have direct or indirect impact on the economics, environment and social parameters of the society. The use of microbes for the production of industrially important metabolites is not a recent event. Mankind has been producing alcoholic beverages and dairy products since the beginning of the civilization but they were oblivious of the role of microbes in the production of these products. The exact role of microbes in the fermentation process was convincingly shown for the first time by Louis Pasteur in 1850s. With the refinement of technology to obtain pure cultures, the association of specific microbes with specific products gained momentum. Table1 lists some of the historical events in the development of industrial Biotechnology.
In 1930s accidental discovery of penicillin by Alexander Fleming boosted the importance of Industrial Biotechnology in the eyes of mankind. The association of an industrially important metabolites with microbes revolutionized the industry to look for novel microbes that produced various kinds of antibiotics, enzymes, anti-cancerous compounds and other chemicals. With the developments in the fields of biochemistry, purification process brought down the cost of recovery of industrial products made with the help of microbes. Industrial Biotechnology received a shot in the arm with the advent of genetic engineering in 1970s. The cloning of genes of industrial importance, from plants and animals into microbes and from one type of microbe e.g. fungus into another genetically stable and highly characterized microbe e.g. E.coli resulted in very high yields of the products at minimal cost. By the use of site directed mutagenesis of genomic DNA, scientists could bring out the permanent change in the gene and hence the gene product. During the last decade, industrial Biotechnology has been revolutionized by the use of high throughput screening, DNA microarray system and bioinformatics.

Microbial Products of Industrial Use

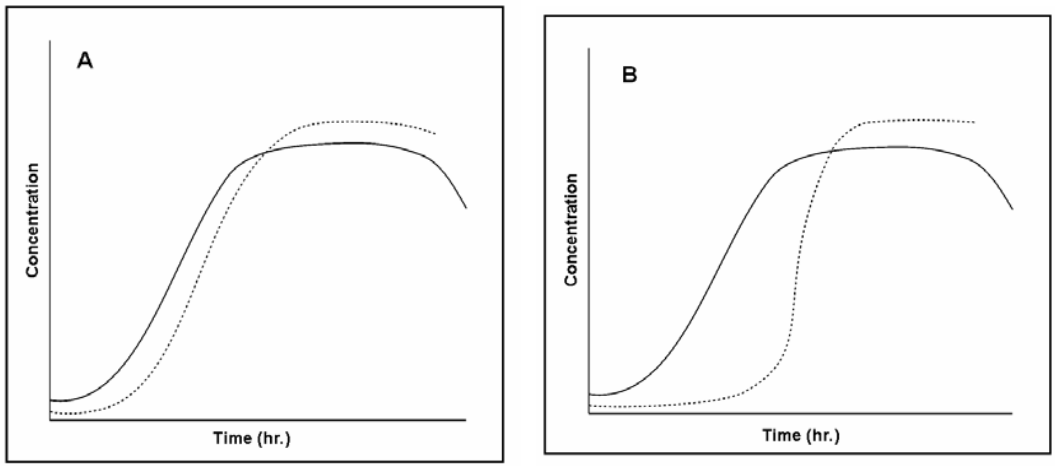
A vast range of industrial products which were earlier made by chemical processes are now being made with the help of microbes. Contrary to the belief that microbes are harmful to humans, majority of microbes are either harmless or provide beneficial products to humans. Microbe based industrial products have made inroads in all walks of life i.e. health sector, food sector, agriculture sector, industrial chemical sector and environmental sector. These products can be categorized into four main groups (Table 2)

Table 2: Microbial products of industrial importance

Category	Product
Primary metabolites	Enzymes, amino acids, nucleotides, organic acids, ethanol, butanediol
Secondary metabolites	Antibiotics, gibberellins, hormones, pigments, alkaloids
Microbial biomass	Baker's yeast, single cell protein (SCP), probiotics, vaccines
Recombinant products	Insulin, streptokinase, interferons, Interleukins, growth hormones, vaccines

Primary metabolites

These are those metabolites which are produced when the cells are growing actively in the presence of sufficient amount of nutrients (Fig. 2). This phase of growth of cells is called **trophophase**. Most of the primary metabolites (amino acids, nucleotides, proteins, nucleic acids, lipids, carbohydrates) are essential or almost essential for the growth and survival of the organisms. Some of the primary metabolites have found industrial applications e.g. ethanol, citric acid, glutamic acid, lysine, vitamins etc.



Growth of cells (-), Metabolite production (----)

Fig. 2: Correlation of microbial growth with production of primary metabolites (A); and secondary metabolites (B)

Secondary metabolites

Secondary metabolites are produced by the cells after the active growth of cells has ceased. This phase of growth of cells is called **idiophase**. Cells do not divide but are metabolically active. Secondary metabolites are not essential for the survivability of the cells. Not all classes of microbes exhibit secondary metabolism. It is commonly seen in fungi, yeast,

actinomyces but is absent in a few bacteria like *E.coli, Salmonella, Shigella, Proteus, Klebsiella* etc. The microbes produce secondary metabolites when their growth rate either slows down or there is complete cessation of growth. Both conditions result in the production of secondary metabolites by microbes. Practically the cells exhibit secondary metabolism when there is insufficient availability of nutrients or suboptimal physical conditions like lack of oxygen or deviation of pH of the surrounding environment from its optimal pH of growth. Such conditions are normally encountered by microbes growing under natural environment. In other words, during natural conditions of growth, it's the idiophase that prevails rather than the trophophase.

In the cells there is correlation between primary and secondary metabolism. Secondary metabolism succeeds primary metabolism (Fig.3). Alternatively, secondary metabolites are produced from intermediates and end products of primary metabolites. A particular secondary metabolite is produced by only a few selective microbes e.g. an antibiotic penicillin is produced only by *Penicillium* spp.

Table 3: Microbial biomass production

Product	Source	Use
Baker's yeast	*Saccharomyces cerevisae*	Production of bread, Single cell protein
Pruteen	*Methylophylus methylotrophus*	Single cell protein
Spirullina protein	*Spirulina maxima*	Single cell protein
Mushroom	*Agaricus bisporus*	Single cell protein
Bioleaching agent	*Thiobacillus ferrooxidans*	Solubiliztion of metals from ores
Bioinsecticide	*Bacillus thuriengensis*	Control of cotton eating insects

Table 4: Recombinants of industrial importance

Gene	Original source	Gene expressed in	Usage
Insulin	Human pancreas	*E.coli, Pichia pastoris*	Control of diabetes
Chymosin	Calf stomach	*E.coli*	Cheese production
Human growth hormone	Human pituitary gland	*E.coli*	Increase in height
Clotting factor VIII	Blood cells	*Saccharomyces cerevisae, Pichia pastoris*	To stop bleeding
Tissue plasminogen activator	Blood cells	*Saccharomyces cerevisae, Pichia pastoris*	Dissolution of blood clots
Streptokinase	*Streptococcus equisimilis*	*Pichia pastoris*	Dissolution of blood clots

Fermentation medium

The growth medium (liquid or solid) in which microbes grow and multiply is called fermentation medium. The selected microbe should be able to utilize and grow on cheap sources of carbon and nitrogen. Usually these sources are waste products of industrial processes e.g. molasses, whey, corn steep liquor etc. Care is taken to avoid the use of such microbes which require expensive nutrients like vitamins for their growth.

Preparation of Culture Media

Objectives

- Be able to introduce of culture media and microbial requirements for the growth.
- Be able to prepare both a complex and selective medium.

I. Introduction

Microorganisms need nutrients, a source of energy and certain environmental conditions in order to grow and reproduce. In the environment, microbes have adapted to the habitats most suitable for their needs, in the laboratory, however, these requirements must be met by a culture medium. Depending on the type and combination of nutrients, different categories of media can be made.

I.1 First: Some basic nutritional requirements must be added in culture media

		Functions
Major nutrients	carbon	component of organic cell materials
	nitrogen	component of proteins, nucleic acids and coenzymes
	sulphur	component of some amino acids and coenzymes
	phosphorus	component of nucleic acids, phospholipids and coenzymes
Macronutrients	potassium	major inorganic cation, cofactor for certain enzymes
	magnesium	important cellular action, cofactor for many enzymes, including those involving ATP synthesis, biding of enzymes to substrates
	calcium	important cellular action, cofactor of some enzymes
	iron	component of cytochromes and some other conjugated
Micronutrients	manganese	cofactor for some enzymes
	cobalt	component of vitamin B12 and coenzymes derived from it.
	copper, zinc, molybdenum	inorganic components of special enzymes

I.2 Second: Categories of culture media

A culture medium is a solid or liquid preparation used to grow, transport, and store microorganisms

1. Complex media: are media that contain some ingredients of unknown composition and/or concentration; this type supplies amino acids, vitamins, growth factors, and other nutrients.
2. Synthetic (defined) media: are media in which all components and their concentrations are known.

I.3 Types of Media

1. A pure culture is a population of cells arising from a single cell.
2. Selective media favor the growth of particular microorganisms and inhibit the growth of others.
3. Enriched media are supplemented by special nutrients to encourage the growth of fastidious heterotrophs.
4. Differential media distinguish between different groups of bacteria on the basis of their biological characteristics.

I.4 Preparation of media

I.4.1 Complex medium for yoghurt bacteria:

Tryptone Glucose Extract agar (T.G.E)

Material

Peptone	10 g
Beef extract	3 g
Lactose	5 g
Agar	20 g
D.W	1000 ml

Procedure:

1. Add 500 ml of distilled water to a Erlenmeyer flask. Weigh out 10 g of peptone, 3 g of beef extract, and 5 g of lactose; dissolve these in distilled water by stirring with a magnetic stirrer and complete this solution into 1000 ml. (Note: Heating may be necessary).

2. Divide this broth solution into two equal parts by adding 500 ml each to two 1000-ml Erlenmeyer flasks:

Flask A: For make broth media.

- Dispense the broth medium (Flask A) into 15 test tubes, adding 9 ml of broth to each tube, then autoclave at 121°c, 1.5 atm for 15- 20 min.
- **Hint:** Adjust pH to 7.2.

Flask B: For make solid media.

- Add 10 g agar (heat with stirring), then adjust pH to 7.2.
- Autoclave at 121°c, 1.5 atm for 15- 20 min.
- After sterilization dispense the nutrient medium into Petri dishes.

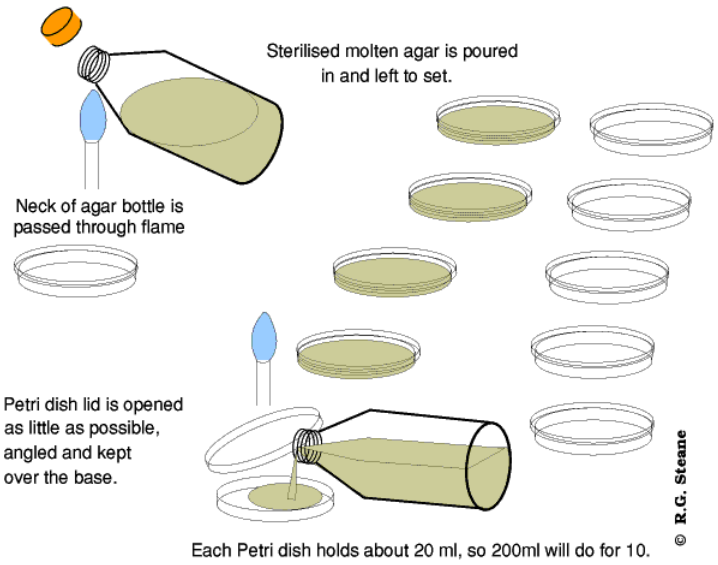


Fig. 1.1 Preparation of media

I.4.2 Yoghurt bacteria

Lactobacillus delbrueckii subsp. Bulgaricus & Streptococcus salivarius subsp. Th mophilus .

I.4.2.1 *Lactobacillus delbrueckii subsp. Bulgaricus*

Lactobacillus delbrueckii are gram-positive, rod with rounded ends 0.5-0.8 by about 2.0-9.0 µ , occurring singly and short chains. Facultatively anaerobic, homofermentative lactic acid bacterium, plays an important role in the dairy industry because of its efficient and reliable utilization of milk constituents, mainly lactose and caseins, and its good resistance to low pH.

1.4.2.2 *Streptococcus salivarius subsp. Thermophilus*

Streptococcus thermophilus are gram-positive, spherical to ovoid cells 0.7-0.9 mm in diameter, in pairs to long chains. Facultative anaerobe that belongs to a group of lactic acid bacteria that convert lactose to lactic acid and acetaldehyde.

Acid is produced from fructose , glucose , mannose ,and lactose. Many ferment sucrose and some ferment maltose weakly.

1.4.3 Selective media for *Lactobacillus delbrueckii subsp. Bulgaricus & Streptococcus salivarius subsp. Thermophilus* .

1.4.3.1 Elliker Agar

Material

Tryptone	20 g
Yeast extract	5 g
Gelatin	2.5 g
Dextrose	5 g
Lactose	5 g
Sucrose	5 g
Sodium chloride	4 g
Sodium acetate	1.5 g
Ascorbic acid	0.5 g
Agar	15 g
D.W	1000 ml

- **Hint:** Adjust pH to 6.8.
- The mixture was then autoclaved, and upon cooling, it was poured into empty petri dishes and solidified.

1.4.3.2 M17 Liquid Media

Material

Phytone peptone	5 g
Polypeptone	5 g
Yeast extract	5 g
Beef extract	2.5 g
Lactose	5 g
Ascorbic acid	0.5 g
2-Disodium glycerophosphate	19 g
MgSO4·7H2O (1.0 M)	1 ml
Glass-distilled water	1000 ml

- **Hint:** Adjust pH to 7.1.
- The mixture was then autoclaved, and upon cooling, it was poured into empty test tubes when needed.

Microbes

These are tiny organisms which require the use of microscopes for their visualization. Microbes are highly versatile organisms. They have many positive features which are responsible for their uses in the field of industrial microbiology.

i) Microbes grow and multiply very rapidly. Billions of cells can be produced in a single day e.g. *E.coli* doubles itself in 15-20 minutes under optimal conditions of growth.

ii) Microbes require cheap nutrients for growth which are available throughout year.

iii) Microbes have great biodiversity. They can grow under extreme conditions of temperature, pH, salts, pressure etc.

iv) The genotype of industrially important microbes is fully known. This aspect helps us to understand the biosynthetic and regulation of a desired gene product. In addition, scientists can either shut off undesirable biosynthetic pathways or boost the yield of desired gene product.

v) Microbes can be stored for years together without any effect on their productivity/desired characters.

vi) Microbial process is eco friendly.

vii) Microbial biomass left after the fermentation can be used as excellent fertilizer.

In addition the selected microbe, chosen for the production of a bioproduct, should have the following properties as well:

Table 1: Landmarks in the development of industrial microbiology

Landmark	Date
Fermentation to produce alcoholic beverages	6000-2000 BC
Vinegar formation	Pre-3000 BC
Production of beer	4000 BC
Visualization of microbes by Leeuwenhoek	mid 17th century
Discovery of fermentation by yeasts	1818
Involvement of microbes in the production of lactic acid	1881
Discovery of fermentation enzymes from yeast	1897
Microbial process for the production of butanol and acetone	1915-1916
Solid substrate fermentation (of citric acid)	1922
Microbial transformations	1937
Commercial production of penicillin	1941-1944
Discovery of other antibiotics	1950
Production of single–cell protein	1960
Use of immobilized enzymes for high-fructose syrup	1960
Commercial use of genetically engineered microbes	1982
Designer enzymes, combinatorial approach	1995
Cloning of secondary metabolite operons	1990
High throughput screening of industrially significant metabolites	2000

i) It should not be pathogenic (except when used for the production of vaccine/toxin) and should not cause any allergic reaction.
ii) It should not lose its desired property on sub culturing.
iii) It should be genetically stable.
iv) It should be amenable to mutation as and when desired.

Safe microbes

For any microbe to be used at commercial level, it is imperative that it should not be pathogenic and the final purified product should be free from toxic contaminants. Prior clearance from regulatory authorities have to be sought for their use at industrial level. However, a few microbes have been categorized as GRAS (generally regarded as safe). No permission from authorities is needed for their use at industrial level, as long as they are grown under stipulated conditions. Organisms listed under GRAS category mentioned .

Microbes classified as GRAS

Bacteria

Bacillus subtilis
Lactobacillus bulgaricus
Lactobacillus lactis
Leuconostoc oenos

Yeasts

Candida utilis
Kluyveromyces marxianus

Kluyveromyces lactis
Saccharomyces cerevisae
Fungi
Aspergillus niger
Aspergillus oryzae
Mucor javanicus
Penicillium roqueforti

Screening of Microbes
Isolating microbes from environment is the first and foremost step in screening for an industrial product which may be either a primary metabolite or secondary metabolite.

Although numerous media have been devised for the growth of microbes, there is no single media that supports all kinds of microbes. For example, if a soil sample is plated on nutrient agar (pH 7.0) and incubated at 37°C for 24-48 hours, only a fraction of aerobic mesophiles will be able to grow. We"ll be missing anaerobes, thermophiles, psychrophiles, alkalophiles and acidophiles. Even slow growing mesophiles (e.g. actinomyces) as well as fastidious mesophiles will not get a chance to grow on such media. Also, if a microbe with antibacterial properties is picked up during screening, it may not allow the growth of some/ all of other mesophiles.
Another factor that is quite important for the selection of microbes depends upon existing capabilities of the scientists as well as the existing facilities of the industrial unit. For example the company undertaking the process may excel in handling the bacteria but not fungi. Therefore, these companies would prefer to isolate bacteria (with desired characteristics) and not fungi or algae.
While choosing a microbe for industrial use, specific considerations have also to be kept in kind. The organism to be used should neither be pathogenic (except in case of production of pathogenic factor, vaccine etc.) nor it should be induce allergic reaction. Also its disposal in the environment (at the end of fermentation) should be eco-friendly.
After narrowing down on a specific product to be extracted from the microbes, the first major query which comes to the mind of a microbiologist is, from where to isolate that specific microbe from the environment. The site for isolation of microbes is decided by the type of microbe one is interested in (Table 10).

Table 10: Preferable sites for isolation of particular microorganisms

Sr. No.	Product	Site
1.	Thermophiles	Hot spring, thermophilic vents, compost
2.	Alkalophiles	High salinity soils and wells
3.	Acidophiles	Pine growing sites, fruits
4.	Osmophiles	Seas and oceans
5.	Yeast	Wine yards
6.	Phosphate solubilizers	Agricultural soil, rhizosphere
7.	Protease producers	Dairy industry, meat processing units.
8.	Enzymes like cellulase, pectinase, xylanase etc.	Agricultural soil, paper and pulp industry, waste disposal area of fruit juice plants, grain market.
9.	Biogas (methane) producers	Gober-gas plants, effluent collection sites.

Isolation of Pure Culture

Objectives

- Be able to introduce of isolation of pure culture.

Introduction

Microorganisms are found throughout the environment: in the air and water; on the surface of any object such as clothes, walls, furniture; in soil and dust; and on and in our own bodies (skin and mucous membranes). Microbes are even found in extreme, seemingly inhospitable environments: living and thriving microbes are common in hot springs with temperatures in excess of 95°C, in Antarctic seawater below 0°C, and in acid mine waters whose pH is below 4, so in nature, microbes rarely occur as pure cultures. Streak plate methods can separate most mixed cultures. Figure 1.1 shows a three – way streak plate. Each streak progressively dilutes the bacteria so that the end of the third streak deposits isolated cells onto the agar. Each cell then divides and grows

into a pure colony or clone. Pure culture technique is a method of culturing microorganisms in which all of the individuals in a culture have descended from a single individual.

Colony isolation:

The basis of pure culture technique is the isolation, in colonies, of individual cells, and their descendants, from other colonies of individuals.

This is usually done by culturing methods employing Petri dishes such as:

1. Streaking.

2. Pouring.

3. Spreading.

Isolation from the wild:

- When isolating microorganisms from complex mixtures it is always a good idea to repeat the isolation procedure at least once (e.g., restreak an isolated colony) to make sure that an isolated colony is truly derived from only a single cell (i.e., closely overlapping colonies can be indistinguishable from colonies founded from single cells).
- Following their isolation from the wild, microorganisms may be characterized by inoculation into differential medium to determine what type of nutrients they require or can use, and what types of by-products they produce. This aids in identification.

I. Streaking

Streaking is a method of applying cultures to solid medium:

1. A sterile loop is cooled and brought into contact with a culture.

2. The loop is then brought into contact with the surface of solid medium whereupon it is streaked (i.e., dragged) along the surface of the solid medium.

3. Colonies grow along the points of the streak. See figure 1.1.

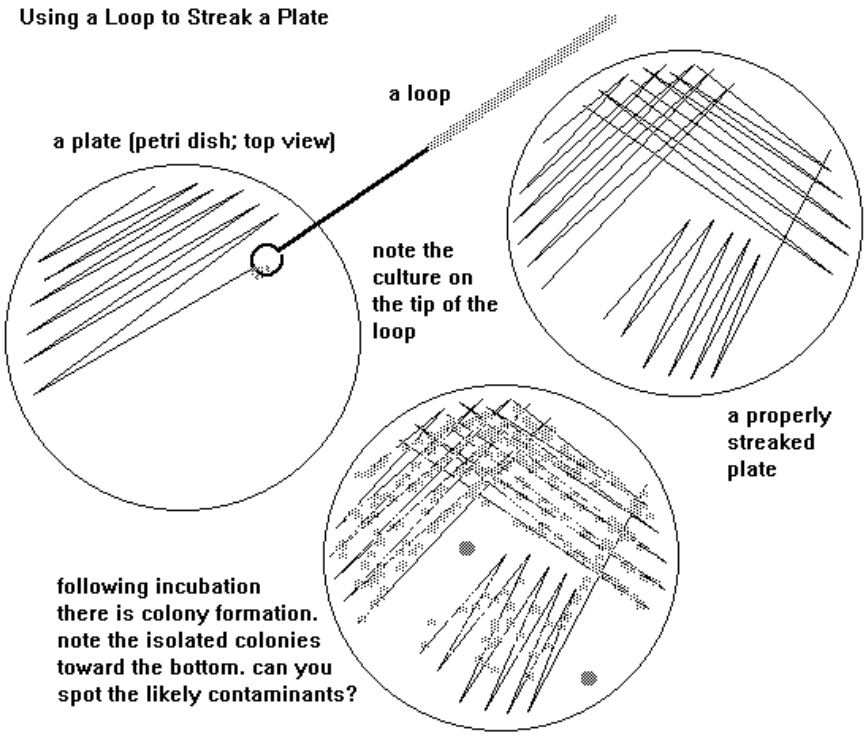


Figure **2.1** Streaking

II. Pouring a plate

A pour plate is a method of melted agar inoculation followed by Petri dish incubation.

1. Cultures are inoculated into melted agar that has been cooled to 45°C.
2. The liquid medium is well mixed then poured into a Petri dish.
3. Colonies form within the agar matrix rather than on top as they do when streaking a plate.
4. Pour plates are useful for quantifying microorganisms that grow in solid medium.
5. Because the "pour plate" embeds colonies in agar it can supply a sufficiently oxygen deficient environment that it can allow the growth and quantification of microaerophiles.

III. Spreading a plate

Quantification technique:

1. Spreading a plate is an additional method of quantifying microorganisms on solid medium.
2. Instead of embedding microorganisms into agar, as is done with the pour plate method, liquid cultures are spread on the agar surface using a devise that looks more or less like a hockey stick.

3. An advantage of spreading a plate over the pour plate method is that cultures are never exposed to 45°C+ melted agar temperatures.

Strain Development Strategy

An ideal industrial strain should have the following qualities:

i) The organism should be able to use wide range of low cost and easily available substrate through out the year.
ii) The organism should produce large biomass and high amount of industrial product.
iii) The organism should be non-pathogenic and non-allergic.
iv) Organism should not degrade with passage of time i.e. it should be genetically stable.
v) It should be easy to perform genetic manipulation on it.

Unfortunately the organism isolated from the environment for industrial use does not fulfill all these requirements. Any property whose addition or deletion from the strain results in minimizing its cost production comes under the purview of strain improvement. It can be:

i) Increase in bio-product formation.
ii) Release of intracellular bio-product by altering permeability of cells.
iii) Overcoming feedback inhibition/repression phenomenon.
iv) Simplifying procedure for extraction/purification of bio-product from

fermentation medium.

Strain improvement by genetic manipulations

In addition to growth media optimizations, genetic manipulation of the organism have also been used for improving the productivity of the strain. The genetic manipulations can be brought about by a mutations and genetic engineering process.

Mutation: The permanent inheritance change in the genetic make up of the organism is called mutation. The agent which brings about the mutation is called mutagen and the organism in which mutation is carried is called a mutant as it possesses mutated gene/genes. In the absence of mutagen the mutation rate is very low i.e.10^{-8} to 10^{-10} per generation/per gene. However the mutation rate in the cells can be enhanced 100 -100,000 folds with the treatment of cells with a mutagen. Basically there are two types of mutagens: a) physical mutagens, e.g. radiations (uv-rays, x-rays, gamma-rays) and b) chemical mutagens, e.g.nitrosomethyl guamidine (NTG), ethylmethane sulphonate (EMS), nitrous acids (NHO_2 etc.).

Protoplast fusion: Protoplasts are single cells from which cell wall has been removed. Protoplast fusion provides a unique tool to combine valuable traits beyond the feasibility of normal sexual crossing. It is widely applied in Plant biotechnology, but is gaining importance in microbes, especially fungi. In this technique, cells from genetically related or unrelated species are fused to produce hybrid cells of desirable traits.

Gene cloning: Gene cloning, as the name suggests, is the cloning of a gene of interest. As the genes of all the organisms (microbes, plants and animals) are made up of same genetic material i.e. nucleotides (adenine, guanine, cytosine and thymine), one can easily shuffle the genes of one organism into another. By means of gene cloning the production of many industrially important bioproducts has been increased many folds. Using this technique, not only the copy number of the concerned gene can be increased but the expression of the gene can be enhanced by replacing the original weak promoter of the gene with a strong promoter. The process of gene cloning is explained below.

Microbial Growth and its Kinetics

The growth of microbes in a liquid medium might result in either homogenous suspension (exhibited by unicellular bacteria) or heterogeneous suspension in the form of flocks, pellets or mats (exhibited by filamentous antinomyces and fungi). The growth kinetics of unicellular bacteria can be calculated by mathematical equation and it is of utmost importance when large fermentors are operated upon. A typical unicellular bacteria exhibit four phases of growth when grown in a liquid medium (Fig. 15). These phases are lag phase, log phase, stationary phase and death phase.

In lag phase, although cells are metabolically active but they do not multiply. The metabolic activity of cells is associated with the adaptability of the organisms to new environment (pH, temperature, heat, oxygen tension, sources of carbon and nitrogen etc.). The organisms synthesize new enzymes required for new metabolic activities and shut off those enzymes or

proteins which are not needed. In log phase, each cell divides into two daughter cells. The cell division occurs with increasing frequency until the maximal growth rate (μ_{max}) is reached. The cell number and cell biomass increases at a constant rate. This exponential growth can be described by two methods; one is related to biomass (x) and another to cell number (N).

The rate of growth of a cell is dependent on biomass concentration and is calculated as follows:

$\mu x = dx/dt$

μx = Specific growth rate / hour.

t = Time in hours

x = Biomass concentration (g/l)

Kinetic Growth Study

Objectives

- Be able to introduce of kinetic growth study.

Introduction

Bacteria grow **FAST!** In ideal circumstances, some common bacterial cells can divide and double **every 20 minutes**. In the following experiment, you are going to use different methods for counting bacteria - **LOTS** of bacteria! Microbiologists use many different ways to "count" bacteria. Some are direct methods, such as counting cells under a microscope ("total count"). Others are indirect methods, e.g. electrical resistance, ATP production).

In this experiment, you will:

1. Use an **indirect** method to count the number of cells in a culture of bacteria (optical density).
2. Use a **direct** method to count the number of cells in a culture of bacteria (viable count).

The Optical Density of Bacterial culture:

The optical density of a bacterial culture is a measurement used by microbiologists to determined the amount of bacterial cells present in a liquid culture. Optical density is the amount of light that is able to pass through a liquid culture. The more bacterial cells in a culture, the denser the culture. This means that less light is able to pass through the sample and the culture is considered to be cloudy to the naked eye.

Spectrophotometer:

A spectrophotometer or Spect 20 is an instrument that measures the amount of light that is able to pass through a bacterial culture. It shines a constant beam of light on the sample that is being tested. If the light hits the bacterial cell, then it will bend and bounce off of the cell. The more cloudy a culture is, the more bacterial cells are present within the culture allowing less light to penetrate through and more light is bounced back to the register within the Spect 20. This will register as a higher value on the machine.

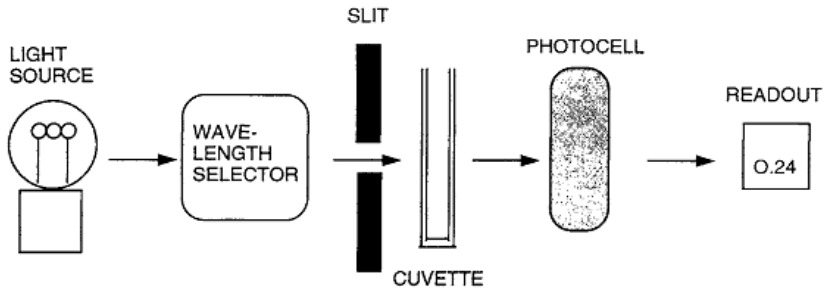


Fig. 5.1 Diagram of components of spectrophotometers.

Materials:

N.B	Incubator
Test tubes	Micropipettes
Flask	Tips
L shape	Cuvettes
Spectrophotometer	

Procedure:

1. From prepared a 10ml culture of *Lactobacillus* sp. and *streptococcus* sp. bacteria. This was grown overnight at 30°C.

2. You will take a 2.3ml sample of the culture and then put them into 23ml of N.B and measuring the optical density of the culture by using of spectrophotometer.

Read the optical density of your sample at a wavelength of 660 nm ($O.D._{660}$). To do this:

a. Adjust the wavelength to 660 nm.

b. Place a cuvette containing a blank of medium in the machine and adjust so the reading is zero.

c. Place a cuvette containing your sample in the machine and read the optical density (O.D.).

Viable Count:

Since you have little idea of the number of viable bacteria in your culture you need to make several dilutions and hope to plate out a dilution giving the appropriate number of colonies.

Procedure:

1. Set up 9 tubes containing 9 ml of Saline solution.

2. Using a pipette with a sterile tip, aseptically transfer 1 ml of your culture into the first tube (10^{-1} dilution). Discard the tip.

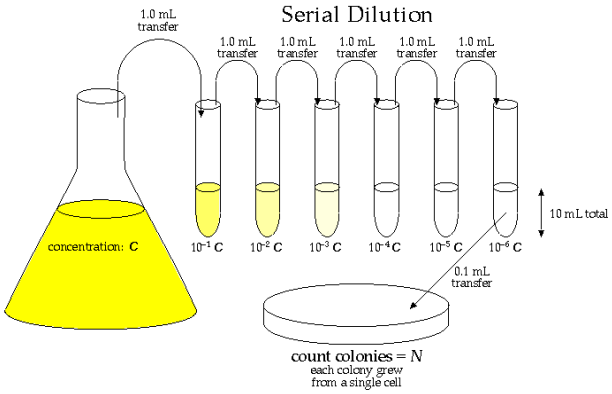
3. Mix well and using a fresh pipette tip, transfer 1 ml to the next dilution tube (10^{-2} dilution). Discard pipette tip.

4. Proceed in this way with your dilutions down to 10^{-9}.

5. Using a fresh pipette tip and starting with the highest dilution, take 0.1 ml of suspension and transfer it to the surface of N.A.

6. Take a spreader and spread out your bacterial dilution over the whole surface of the plate.

7. Once dry, place the plates in the incubator for incubation at 30°C.



Result for dilution down to 10^{-6}:

10^{-1} dilution	
10^{-2} dilution	
10^{-3} dilution	
10^{-4} dilution	
10^{-5} dilution	

10^{-6} dilution	

Hint: For every dilution you can read the optical density of your sample at a wavelength of 660 nm ($O.D._{660}$) and make some calculation or plot the result using (Growth curve) as the following:

First: Calculations

- Total number of bacteria = number of colonies × dilution factor × 1/ volume

Therefore, bacterial counts by these methods are usually expressed as colony forming units per milliliter (CFU/ ml).

- The increase in cell numbers that occurs in an exponentially growing bacterial

  culture that occurs is a simple geometric progression of the number 2. As two cell double (to become four cells), we can express this as $2^1 \rightarrow 2^2$. As four cells become eight, we express this as $2^2 \rightarrow 2^3$, and so on. Because of this geometric progression, there is a direct relationship between the number of cell present in a culture initially and the number present after a period of exponential growth:

$$N = N_0 2^n$$

Where N = the final cell number, N_0 = the initial cell number and n = the number of generation.

- From this equation we can find the number of generation (n) by taking the log.

$$\log_{10} N = \log_{10} N_0 + n \log_{10} 2$$

or alternatively:

$$n = (\log_{10} N - \log_{10} N_0) / \log_{10} 2$$

Hint: (n in some texts express as x = number of generation).

- The mean generation time or "doubling time" **(g)** or (t_D) is the average time required for all the components of the culture to double. This is calculated from the following equation:

$$t_D = T / n$$

Where t_D = doubling time, T = total elapsed time and x = number of generations.

- Another parameter used to describe cell growth is specific cell growth rate (μ). Specific cell growth rate (μ) is a measure of the rate of cell number increase at a particular cell concentration.

- The value of μ can be determined from the following equation:

$$\ln N_t - \ln N_0 = \mu(t - t_0)$$

For most purposes, it is easier to use $\log_{10}$ values rather than natural logs, so the above equation can be converted as follows:

$$\log_{10} N - \log_{10} N_0 = (\mu/2.303)\ (t - t_0)$$

or alternatively:

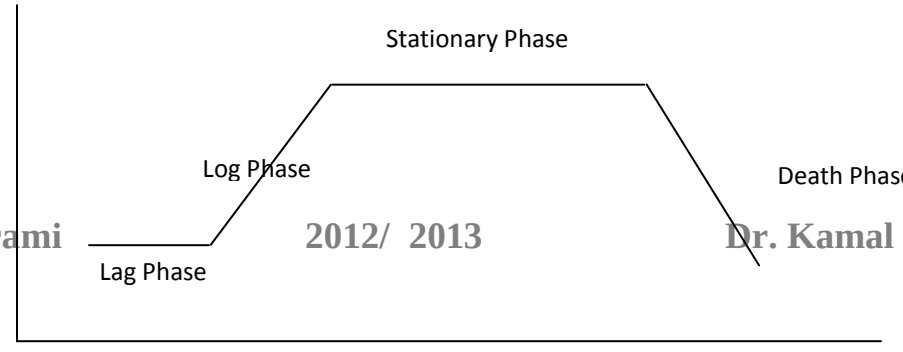
$$\mu = (\ (\log_{10} N - \log_{10} N_0)\ 2.303) / (t - t_0)$$

- If the doubling time is already known the specific growth may also be calculated using the following formula.

$$\mu = \ln 2 / t_D$$

Second: Graph Bacterial Growth

Using semi-log paper, OD values from the spectrophotometer are combined to show a growth curve with respect to time. This is an example of what a growth curve looks like.



OD^{660}

Time

1. During the **LAG PHASE**, the bacteria in the original inoculum's adapt to their new environment (e.g. synthesis enzymes required for growth). The number of cells does not increase during this time (and under some circumstances may even decrease slightly). The lag phase is longest when there is the greatest difference between the previous and present growth conditions of the inoculum, e.g. when stationary phase cells are transferred to fresh medium or when cells in a rich medium are transferred to a nutrient-poor medium

2. During the **EXPONENTIAL or LOGARITHMIC PHASE** of growth, the number of bacteria doubles at regular intervals. During the exponential phase, "balanced growth" occurs, i.e. a balanced increase in all cellular components.

3. During the **STATIONARY PHASE**, the bacteria have exhausted one or more critical requirements for growth. Growth slows as wastes accumulate and/or nutrients are depleted. Bear in mind that a bacterial culture is a dynamic situation - individual cells are dividing or dying all the time. During the stationary phase, the rate of cell division is equal to the rate of cell death, thus the number of viable cells remains constant. Cells in the stationary phase have significantly different properties from those in the exponential phase of growth.
4. During the **DEATH or DECLINE PHASE**, the rate at which cells die is greater than the rate at which they divide - hence the number of viable cells decreases. Note that there is a exponential decline in the number of cells during the death phase - the reciprocal of the logarithmic phase.

Fermentor & Fermentation Process

A fermenter, also called a bioreactor, is a vessel in which a particular microbe is grown under controlled conditions to produce a desired byproduct or biomass. The aim of the fermenters is to provide a stabilized condition for growth of cells and better production of a desired byproduct. In an ideal fermentation, important parameters of fermentation (pH, temperature, gas transfer, heat transfer, agitation and foaming) are tightly regulated.
In the laboratory, fermentation is carried out in conical flasks as the volume of fermentation medium is small i.e. up to 500-1000 ml. These flasks are made up of glass or plastic material. The flasks are kept in an incubator–shaker in which the temperature and aeration can be regulated.
There are many types of fermenters, but a conventional fermenter (Fig. 16) has the following features: A fermenter is made up of non-corrosive cylindrical steel vessel which can easily withstand a temperature of 150-190$_o$C and a pressure exceeding 380 kPa. The latter conditions are necessary for satisfactory sterilization of the fermenter. The vessel has many inlets for addition of gases, media components, acids/alkalies as well as outlets for escape of gases and sample collection for various tests to be performed during incubation. The vessel also has provisions for ports for various sensors like pH, temperature and dissolved oxygen (DO). The fermentation vessel has a provision for cooling the fermentation broth, if the incubation temperature rises above the requisite temperature. The latter phenomenon is observed in exothermic fermentations like ethanol fermentation. The fermenter is cooled down by passing cold water in the jacket surrounding the vessel or in the coils present inside the vessel. The fermenters are fitted with an agitator for uniform mixing of the media
37
components and gases, maintaining constant temperature as well as uniform suspension of cells in the fermentation medium.

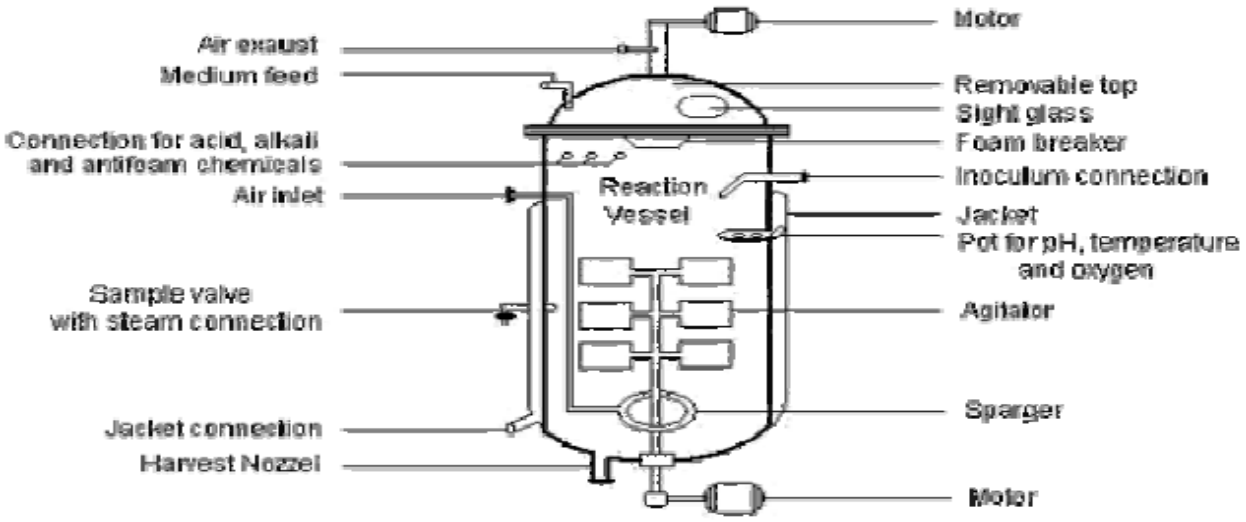
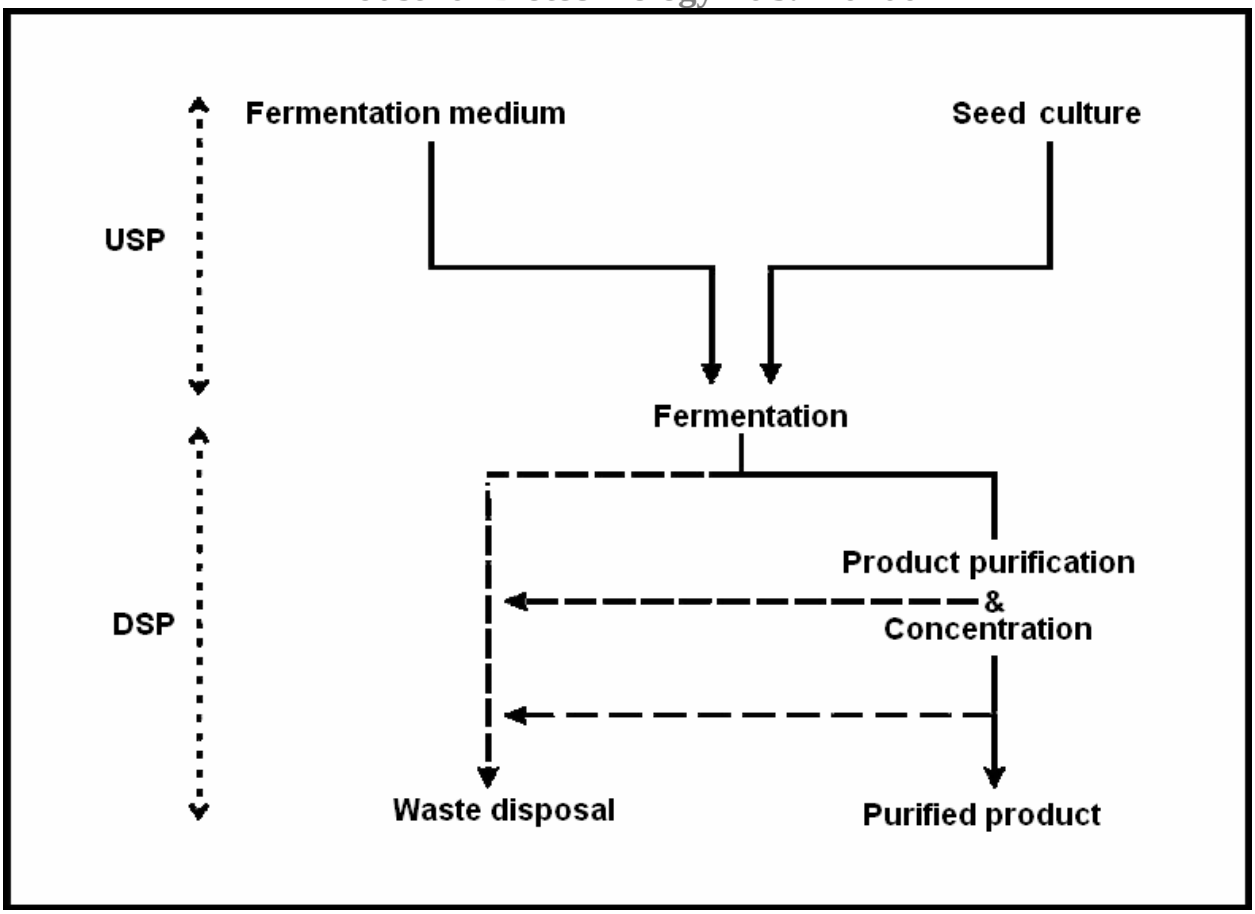


Fig. 16: A typical bioreactor

Fermenter

For production of a desired microbial product, it is of utmost importance to optimize physical (pH, temperature, aeration etc.) and chemical (carbon, nitrogen, mineral sources etc.) composition of the fermentation medium. To maintain these stringent conditions, microbes are grown in containers called as fermenters or bioreactors. The capacity of bioreactors may vary from 10 liters to 100,000 liters depending on the byproduct.

In general, fermentation process is divided into two parts i.e. Up Stream Processing (USP) and Down Stream Processing (DSP, Fig.1). USP is basically involved in the a) preparation of a seed culture (i.e. inoculums preparation), b) preparation of fermentation medium and its sterilization and c) the fermentation process for optimal production of a metabolite. DSP is concerned with the recovery of product, its concentration and purification and disposal of biowaste generated out of the fermentation process.



USP: Up stream processing; DSP: Down stream processing

Fig. 1: Diagrammatic representation of a fermentation process

Table 6: Applications of microbes in healthcare products

Product	Source	Use
Penicillin	*Aspergillus chrysogenum*	Antibacterial activity
Cephalosporin	*Cephalosporium acremonium*	Antibacterial activity
Bacitracin	*Bacillus subtilis*	Antibacterial activity
Streptomycin	*Streptomyces griseus*	Antibacterial activity
Erythromycin	*Saccharapolyopora erythraea*	Antibacterial activity
Gentamycin	*Micromonospora purpurea*	Antibacterial activity
Amphotericin B	*Streptomyces nodosus*	Antifungal activity
Nystatin	*Streptomyces noursei*	Antifungal activity
Lentinan	*Lentinus edodes*	Anti tumour activity
Funiculosin	*Penicillium funiculosum*	Antiviral activity
Cyclosporin	*Cylindrocarpum lucidum Tolypocladium inflatum*	Immunosuppresent
Zaragozic acid	*Sporomiella intermedia*	Cholesterol inhibitor.
Prednisolone	*Arthrobacter simplex*	Biotransformation of hydrocortisone into Prednisolone
11α-	*Rhizopus nigricans*	Biotransformation of progesterone to 11 α-

Table 7: Applications of microbes in food and beverage fermentations

Product	Source	Uses
Enzymes for alcoholic beverages like beer, wine , whisky:		
α-amylase	*Bacillus, Aspergillus*	starch hydrolysis
β-amylase	*Bacillus, Streptomyces, Rhizopus*	starch hydrolysis
Glucoamylase	*Aspergillus, Rhizopus*	starch hydrolysis
Protease	*Aspergillus niger*	Prevention of haze formation
Glucose oxidase	*Aspergillus niger*	Removal of oxygen from beer
Acetic acid (vinegar)	*Acetobacter aceti, Gluconobacter oxydans*	Conversion of ethanol into acetic acid
Yoghurt	*Streptococcus thermophilus, Lactobacillus delbrueckii*	Production of organic acids esp. lactic acid
Cheese	*Streptococcus thermophilus, Lactobacillus, Lactococcus, Propionobacterium, Penicillium*	Production of organic acids esp. lactic acid, flavour enhancers and stabilization of cheese.
Probiotics	*Lactobacillus, Saccharomyces cerevisae, Bifidobacterium*	Stimulate immune system, Stabilizes gut microflora.
Bread	*Saccharomyces cerevisae,*	Rising of bread
Sauerkraut	*Lactobacillus, Leuconostoc plantarum*	Lactic acid production

Types of bioreactors

Based on the design of bioreactors, they can be classified as: a) Stirred tank bioreactor, b) Bubble column bioreactor, c) Airlift bioreactor, d) Fluidized bed bioreactor and e) Packed bed bioreactor

a) Stirred tank bioreactor

It is also called stirred tank reactor (STR). It is fitted with impellors (agitators) which are connected to a motor driven shaft located centrally in the reactor.

b) Bubble column bioreactor

This type of bioreactor lacks impellers. The air is pumped into the fermenter through a plate located at the base of the bioreactor. This plate has many openings through which air or gas is passed in the form of bubbles into the liquid medium contained in the bioreactor. The introduction of air/gas in this manner results in uniform mixing of the fermentation broth.

c) Airlift bioreactor

In this the inside of the bioreactor is divided into two zones, riser zone and downcomer zone, with the help of a baffle or a draft tube. The air enters the vessel through the bottom part of the rizer zone resulting in upward movement of the medium. The upper portion of medium then enters the downcomer zone and flows down to the base of the bioreactor.

d) Fluidized bed bioreactor

In this type of bioreactor the immobilized cells or enzymes, present in the form of beads, are

suspended in the lower cylindrical portion of bioreactor. The air is pumped in the vessel from the bottom of the bioreactor. The upper portion of the bioreactor comprises of cell free broth containing the bioproducts of fermentation. This upper portion is collected in a separate tankas the new medium is pumped in the lower vessel.

e) Packed bed bioreactor

In this type of reactor, the vessel is loosely packed with matrix to which a biocatalyst (cells or enzymes) has been immobilized. The substrate is poured onto the matrix from the top of reactor and the bioproducts are collected from the bottom of the reactor. In a few cases, the substrate is pumped into the reactor from the bottom and bioproduct is collected from the top.

Fermentation processes

There are two groups of fermentation processes: liquid fermentation and solid fermentation. In former case, cells are suspended in aqueous medium where as in latter case, the volume of free liquid is minimal and the cells are adsorbed to a solid and nutrient rich material.

Liquid fermentation

There are many forms of liquid fermentation:

i) Batch Fermentation

The culture is inoculated into the sterile medium contained in a closed vessel. No additional nutrients are added once the fermentation process starts. Physical parameters (pH, temperature and aeration) of the fermentation medium are regulated in this type of fermentation.

ii) Fed Batch fermentation

It is quite similar to Batch fermentation except that the nutrients or one or two components of nutrients are added periodically in the fermentation medium. This type of fermentation lengthens the log and stationary phase of the cells thereby causing increased amount of bioproduct.

iii) Semi continuous fermentation

In this type of fermentation, a fixed volume of fermented medium is taken out from the fermenter and the same volume of nutrients is added so as to keep the volume of fermentation medium same and replenishing the nutrients for growth of microbes. This process also results in maintaining the microbes in same phase of growth.

iv) Continuous fermentation

In this case, there is continuous removal of culture medium as well as continuous addition of sterile nutrient medium. Conditions are predetermined as to what should be the flow rate of incoming nutrient solution so that the volume of fermenting medium remains the same and also fermenting microbes remain in same phase of growth termed as steady state of growth. The latter condition can be achieved by keeping constant concentration of carbon source or nitrogen source or dissolved oxygen content. Such type of fermenter is called a chemostat. The steady state of growth can also be maintained by constantly monitoring the turbidity of the culture medium by use of turbidometer. Such type of fermenter is called a turbidostat.

Solid State Fermentation (SSF)

In some cases microbial growth is carried out on the solid substrates. The amount of free water is almost non existent. The microbes adhere to the solid substrates which are source of

nutrients for them. The source of solid material as well as culture is cereal grains of wheat, rice, maize etc. The grains are moistened with water and ground to form a paste. Additional supplements like salts etc. may be added to the solids prior to sterilization. The solid material is then transferred to shallow metallic containers and is steam sterilized. This is followed by the spraying of culture inoculum on to the surface of sterilized medium and incubation is carried out under controlled conditions of temperature, air and humidity. However in some cases like idli, cada, dosa, pickles and jalebi no sterilization is carried out before fermentation. Only when solid fermentation using tray fermentors are used in enzyme production or any such product and pure cultures are used for fermentation then steam sterilization of medium before fermentation is done. Nearly all domestic fermented foods are prepared by use of SSF e.g. idli, vada, pickles and jalebi.

Inoculum Development

The process adopted to produce a culture volume sufficient for inoculation in the fermentation medium is called inoculum development. In order to achieve maximal yield of product via fermentation, the culture inoculum should have the following characteristics: it should be a) metabolically highly active, b) easy to prepare in large volume, c) of suitable morphological form and d) free from microbial contamination.
Ideally, the size of inoculum varies from 2-10% (v/v) of the fermentation medium. Normally for bacterial fermentations inoculum size varies from 2-5% and for fungal fermentations it varies from 3-10%. In industry, the size of fermentation medium can be very high e.g. 100,000 litres, which means that the minimal inoculum size will be 2000 litres. To prepare an inoculum of this magnitude is not an easy task. Starting from a stock culture, which may be in lyophilized form or on a slant-agar, the inoculum is built up in a number of stages. In the first stage, a small amount of culture is inoculated in a shake-flask and incubated. Once the culture has grown, it is transferred to a larger flask and incubated. The inoculum thus prepared is then transferred to a small laboratory fermenter. The size of culture inoculum is further increased by transferring this culture into a pilot scale fermenter. Usually, the chemical composition of seed culture medium and fermentation medium is quite similar so as to minimize lag phase of culture growth.

The inoculum developed using unicellular bacteria or yeast is of uniform consistency. However, filamentous organisms, like actinomyces and especially fungi, the growth of vegetative inoculum is either pellet-type which consists of a compact discreet mass of hyphae or filamentous-type in which fungal hyphae form a uniform suspension in the medium. Filamentous-type of growth results in the formation of viscous medium which is difficult to aerate whereas pellet-type growth give rise to less homogenous type of growth and the cells located in the center of pellet are not metabolically very active. These morphological forms of fungal growth have a great influence on the yield of bioproduct. Filamentous form of inoculum is preferred for penicillin production by *Penicillium chrysogenum* and pellet form is preferred for citric acid production by *Aspergillus niger*.

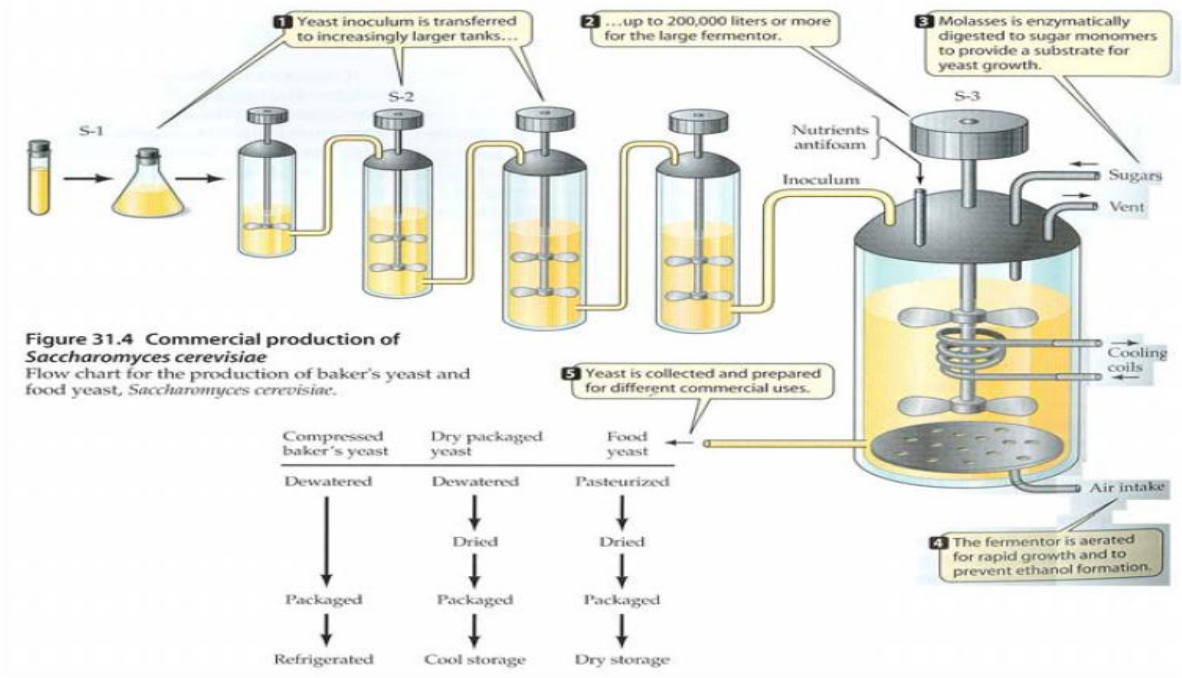


Figure 31.4 Commercial production of *Saccharomyces cerevisiae*
Flow chart for the production of baker's yeast and food yeast, *Saccharomyces cerevisiae*.

Downstream Processing

The extraction and purification of a biological product from the fermentation broth is called as down stream processing (DSP) or product recovery. DSP is quite complex and variable depending upon the type of the product. Broadly it can be divided into five stages:
i) Cell harvesting
ii) Lyses/breakage of cells
iii) Concentration
iv) Purification

v) Formulation

Cell harvesting

Once the fermentation is complete, the solid phase (i.e. cell bio-mass) is separated from the liquid phase by any of the following methods: Flotation, Flocculation, Filtration and Centrifugation.

Flotation:

A gas is passed through the fermentation broth. This gas forms tiny bubbles to which cells get adsorbed. These bubbles rise to the surface of the broth and form a foam. Some times along with the gas, collector substances (fatty acids) are added which facilitates foam fermentation. The foam containing the cells is removed.

Flocculation:

At high cell density some cells (yeast cells) aggregate and thus settle down at the bottom of the fermenters. This process can be accelerated by the addition of flocculating agent like salts, organic polyelectrolyte and mineral hydrocolloid.

Filtration:

It is the most common type of cell separation technique. Many types of filters are used: ***Depth filters:*** These are composed of filamentous materials e.g. glass wools, asbestos or cellulose papers. Cells get entrapped in the matrix and clear cell free material passed out. This technique is used for the separation of filamentous fungi. ***Asbestos filters:*** These filters have specific porosity which can retain cells and let go liquid material. This is most suited for separation of bacteria. ***Rotary drum vacuum filters:*** This is usually applied for the separation of yeast cells. This unit consists of a rotary drum which is partially immersed in a fermentation broth. Rotating drum adsorbs cellular bio-mass. Once the biomass is out of fermentation broth, it is partially dried and then removed from the drum.

Centrifugation: This is a process of separating cells from the liquid based on the differences in their density. The cell culture is spun at a high speed in an instrument called centrifuge. This results in settling of denser material i.e. cells at the bottom or sides of the holder. In industry, continuous flow centrifuges are used in which culture is continuously supplied to the centrifuge. The cells get collected in the centrifuges and cell free supernatant flows out of the centrifuge.

Lyses/breakage of cells

If the desired product is located inside the cell, then the cells are fist recovered from the fermenter by any of the methods explained earlier in this chapter. These cells are suspended in buffer and their cell walls are disintegrated, so as to break open the cells by physical or chemical means in order to release intracellular products in the medium.

1-Physical methods: ***a) Ultrasonication:*** The cells are disrupted by passing ultra-waves through samples. This technique is ideal in laboratory where sample size is small. ***b) Osmotic shock:*** The cells are suspended in a viscous solution like 20% (w/v) sucrose or glucose. The cell suspension is then transferred to the cold water (4°C) which results in cell lyses. ***c) Heat shock (Thermolysis):*** The cells are exposed to heat which results in disintegration of the cells. It is an economical method but the product has to be heat stable. ***d) High pressure***

homogenization: The cell suspension is forced to pass through a narrow pore at a high pressure which results in breakage of cells. ***e) Grinding with glass beads:*** The cell suspension containing glass beads is subjected to a very high speed in a vessel. The cells break as they are forced against the walls of the vessel by the beads.

2- Chemical method: Various chemicals can destroy the cell walls/cell membrane of the cells. These chemicals can be alkali (sodium hydroxide, water miscible organic solvents like alcohol and detergents (sodium lauryl sulphate, triton X-100, cetyl trimethyl ammoniumbromide

Enzymatic lysis: The cells are broken with the help of enzymes. Bacterial cells are lysed by the addition of lysozymes. This enzyme hydrolyses β-1,4 glycosidic linkage of peptidoglycan layer of bacterial cell wall. Fungal cells are lysed by the addition of chitnases, cellulases and mannases

.

Concentration

More than 90% of the cell free supernatant is water and the amount of desired product is very less. Evaporation process concentrates the requisite product. Water is evaporated by applying heat to the supernatant with/without vacuum. The amount of heat applied is decided by the heat stability of the product. Various forms of evaporators are used e.g. plate evaporator, falling film evaporator, forced film evaporator and centrifugal forced film evaporator.

Liquid-liquid extraction: A desired product (solute) can be concentrated by the transfer of the solute from one liquid to another liquid. This process also results in partial purification of the product. Liquid-liquid extraction is basically of two types: (a) extraction of low molecular weight products (b) extraction of high molecular weight products.

Extraction of low molecular weight products: This is primarily applied for extraction of lipid soluble compounds. This process can be carried by the following techniques: a) Physical extraction: The aqueous solution containing the product is mixed with an organic solvent. The amount of product getting into organic solvent depends on the partial coefficient of the product. b) Reactive extraction: It is like physical extraction procedure except that the product is first reacted with a carrier compound (e.g. phosphorus compound, aliphatic amine) and then mixed with organic solvent.

Extraction of high molecular weight: This technique is applied to compounds like proteins, which get inactivated in the presence of organic solvents like ethanol, methanol, chloroform, hexane etc. These compounds are extracted by using an aqueous two phase system (ATPS). In this technique a polymer (e.g. polyethylene glycol) and salt solution (e.g. ammonium sulphate) are added to the aqueous solution containing the product to be concentrated. The particulate matter i.e. cells remain in one phase and the proteins (the desired matter) are transferred to other phase. The solubility of the desired product is dependent on various properties of the protein e.g. ionic character, hydrophobicity, size etc.

Membrane filtration: This technique involves the use of semi-permeable membrane. Many kinds of membranes are available based on the size of the pores and their chemical nature (polyether sulfone or polyvinyl difluoride). Unfortunately, these filters cannot be sterilized. Now membranes made up of ceramics and steel are being introduced which can be easily autoclaved. Membrane filtration techniques like macro filtration, ultra-filtration and hyper filtration

product can combine specifically once the aqueous solvent, containing the product, is passed

through this. The adsorbed material is then eluted using various buffers and salts.

Precipitation: This is the most commonly used procedure for concentration of compounds especially proteins and polysaccharides. The agents commonly used in the process of precipitation are neutral salts (ammonium sulphate), organic solvents (ethanol, acetone, propanol), non-ionic polymers (PEG) and ionic polymers (polyacrylic acid, polyethylene amine).

Purification by chromatography

Chromatography is a procedure for separating molecules based on their sizes, charge, hydrophobicity and specific binding to ligands. The chromatography techniques used are as follows:

Gel Filtration chromatography (size exclusive chromatography): The matrix is made up of tiny beads having many pores in them. Many types of beads are available having different porosity. Small molecules enter the beads whereas large molecules cannot enter and therefore come out of the column first. By this technique protein of variable sizes can be purified.

Ion exchange chromatography: Most of the proteins have a net positive or negative charge. This property of the proteins is exploited for the purification of proteins by passing protein solutions through columns of charged resins. Two types of resins are used in the industry. Cation exchange resins and anion exchange resins. Cation exchangers (carboximethyl cellulose) have negative charged groups whereas anion exchangers (diethyl aminoethyl) have positive charged groups. Proteins carrying net positive charge bind to cation exchangers where as proteins carrying net negative charge bind to anion exchangers.

Affinity chromatography: In this procedure, the proteins are separated based on their affinity for a product compound i.e. ligand .Once the protein is bound to the affinity matrix, it is eluted by changing the pH of the eluting buffer or alteration of ionic strength etc.

Commercial Production of Citric Acid (solid state fermentation)

Citric acid is a component of TCA cycle, also called Krebs cycle. It has immense application in the industrial sector. It is widely used as an acidulant and flavouring agent in food and beverage industry. In chemical industry it is used as an antifoam agent and maintaining metals in solution for electroplating. In pharma sector a salt of citric acid, trisodium citrate is used as blood preservative. Citric acid is also used as an alternative to polyphosphates in detergent industry. The most commonly used strain for citric acid production is an aerobic fungus, *Aspergillus niger*.

Citric acid biosynthesis

Citric acid is a primary metabolite product and is produced by the involvement of Embden-Meyerhof-Parnas (EMP) pathway and TCA cycle. Glucose is first converted to pyruvate. The latter metabolite is decarboxylated to oxaloacetate, which then combines with acetyl CoA to give rise to citric acid. In strains used for citric acid production, TCA cycle is kept active by the involvement of anaplerotic reactions.

Production medium

Easily degradable carbon sources like glucose or sucrose are ideal for citric acid production. For this reason, cane molasses or beet molasses are used. Starch is another source of carbon for citric acid production. For nitrogen source, ammonium ions or urea is added to the medium. Certain trace metals like manganese, iron and zinc are ideal for the growth of *A. niger*. Oxygen is an integral component as *A.niger* is an aerobic organism. The pH of the medium should be less than 3.0 for maximal production of citric acid.

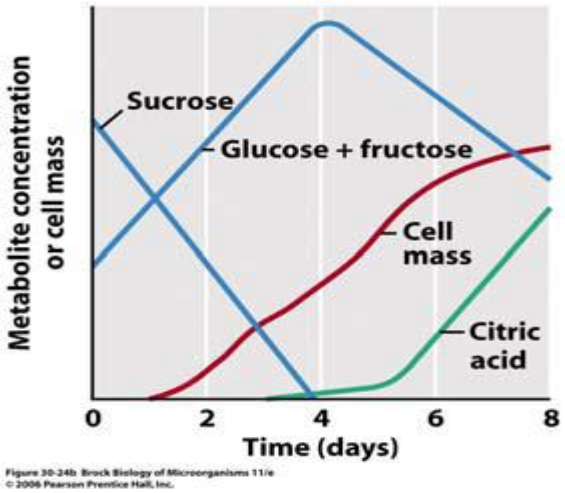


Figure 30-24b Brock Biology of Microorganisms 11/e
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Commercial production

There are two processes by which citric acid is produced in the industry: surface process and submerged process.

Surface process: This process is the very ancient and is in use in developing countries because it is simple and easy to use. In this process the fungus is grown on the surface of the growth medium, which may be solid or liquid.

Solid surface process: The wheat bran or sweet potato waste adjusted to 70-80% moisture content and pH around 4.0 is sterilized in shallow pans and then sprayed with spores of *A.niger*. Incubation temperature is around 28$_o$C. Sterile air is passed through the incubation chamber for adequate supply of oxygen. Optimal yields of citric acid are produced after 70-100 hrs of incubation.

Liquid surface process: The liquid fermentation medium basically comprises of beet molasses supplemented with salts and pH adjusted to 4.0 and sterilized. This medium is transferred to a sterile shallow aluminum or steel pans and inoculated with spores of *A.niger* and incubation temperature of 28$_o$C is maintained. Within 24-48 hrs a layer of vegetative mycelium is formed on the surface of medium. Optimal yields of citric acid are produced after a week to 10 days of incubation.

Submerged process: Nearly 80% of citric acid production is carried out by submerged process. Here, two types of bioreactors are used: stirred tank and air lift bioreactors. The fermentation vessel is made up of high quality stainless steel as citric acid is very corrosive in nature. The media composition is essentially the same as mentioned in liquid surface process. However, vegetative mycelium and not spore suspension is used for the inoculation of the medium. The vegetative inoculum is prepared by inoculating spores in a small liquid medium. After incubation, the inoculum is sub cultured into a fresh and large volume

medium. This process is repeated till desired amount of inoculum in the form of fungal pellets is achieved. Care is also taken to produce pellets which should be small in size (less than 1mm diameter) and have fluffy centers.

Recovery of citric acid

The process for recovery and purification of citric acid is quite similar for surface and submerged fermentations. Initially the solid biomass is filtered out of the aqueous medium. The medium is acidified to pH 3.0 with lime (CaO) to precipitate undesirable co-product, oxalic acid. The acidic pH of broth is neutralized to pH 7.0 and heated up to 70-90$_o$C. This step results in precipitation of citric acid. The precipitates are recovered and treated with dilute sulphuric acid solution. This results in formation of citric acid and precipitation of calcium sulphate (gypsum). After filtration, the dilute solution of citric acid is passed through columns of activated charcoal and ion exchangers so as to remove colours and other impurities. Purified crystals of citric acid are obtained after evaporation of water.

Microbial Biomass Production: Bakers yeast

Production of Yeast and Yeast Extract

Objectives

- Be able to produce of yeast (*Saccharomyces cerevisiae*) by fermentation method.
- Be able to produce of yeast extract.

Introduction

Two main types of baker's yeast are produced, compressed (cream) yeast and dry yeast. The first stage of yeast production consists of growing the yeast from the pure yeast culture in a series of fermentation vessels. The yeast is recovered from the final fermentor by using centrifugal action to concentrate the yeast solids. The yeast solids are subsequently filtered by a filter press or a rotary vacuum filter to concentrate the yeast further.

Yeast cells are grown in a series of fermentation vessels. Yeast fermentation vessels are operated under aerobic conditions (free oxygen or excess air present) because under anaerobic conditions (limited or no oxygen) the fermentable sugars are consumed in the formation of ethanol and carbon dioxide, which results in low yeast yields.

For production:

I.a For 1L of nutrient broth you need:

Materials:

Fermentor	
Saccharomyces cerevisiae	
Sucrose	3- 5%
$MgSo_4.7H_2O$	0.5 g
$(NH4)_2So_4$	5 g
KH_2Po_4	1 g
Yeast extract	1 g

Adjust pH to (4.5- 5)

Procedure:

1. You will prepare from I.a 1000 ml.

2. Before sterilization take 100 ml from I.a and put them into another flask (I.a.1) then autoclave the two solutions at 121°c, 15 atm for 15- 20 min.

3. For I.a.1 add 1g of baker's yeast aseptically after cooling and then Incubate the solution in the incubator at 30°c for 24 hr for activation.

4. Add 100 ml of nutrient broth to 900 ml of nutrient broth aseptically.

5. Finally incubate the whole solution at 37°c for one week.

6. Measuring the optical density of the culture by using of spectrophotometer twice daily then graph yeast Growth.(review lab No. 5)

7. Detection the weight of yeast by using eppendorf tubes as the following:

- Detect the weight zero of the eppendorf tubes.
- Put 1 ml of the fermented solution into it.
- Centrifugation.
- Leave the supernatant.
- Put the eppendorf tubes into the oven at 70ºC for 24 hrs.
- Detect the final weight after 24 hrs.
- Repeat these procedure daily.

- Graph yeast Growth by using the two parameter changeable (weight and time).
- Compare between two graphs.

II. Harvesting and Packaging

Once an optimum quantity of yeast has been grown, the yeast cells are recovered from the final trade fermentor by centrifugal yeast separators. The centrifuged yeast solids are further concentrated by a filter press or rotary vacuum filter. This filter cake is then blended in mixers with small amounts of water, emulsifiers, and cutting oils to form the end product. The final packaging steps, as described below, vary depending on the type of yeast product.

In compressed yeast production, emulsifiers are added to give the yeast a white, creamy appearance and to inhibit water spotting of the yeast cakes. A small amount of oil, usually soybean or cottonseed oil, is added to help extrude the yeast through nozzles to form continuous ribbons of yeast cake. The ribbons are cut, and the yeast cakes are wrapped and cooled to below 8°C (46°F), at which time they are ready for shipment in refrigerated trucks.

In dry yeast production, the product is sent to an extruder after filtration, where emulsifiers and oils (different from those used for compressed yeast) are added to texturize the yeast and to aid in extruding it. After the yeast is extruded in thin ribbons, it is cut and dried in either a batch or a continuous drying system. Following drying, the yeast is vacuum packed or packed under nitrogen gas before heat sealing. The shelf life at ambient temperature is 1 to 2 years.

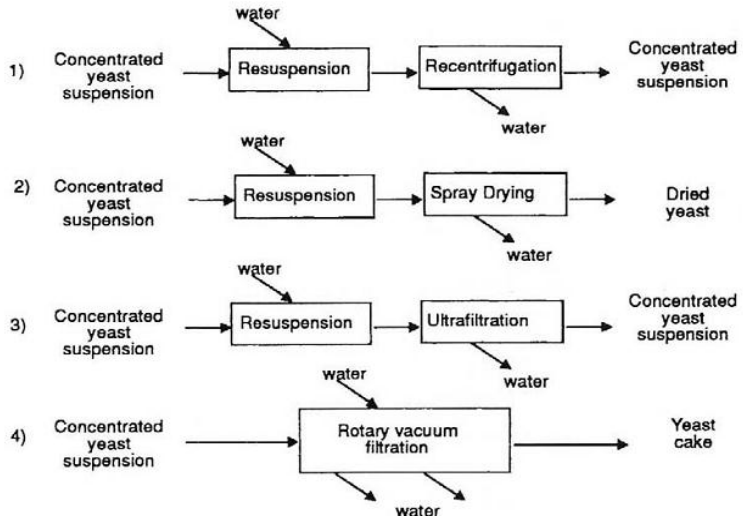


Fig. 6.1 Various Methods for Packaging Yeasts

III. Yeast extract production

Yeast extract is commercially available as either powders or pastes and has been extensively used as flavoring agent by the food industry.

There are three distinct manufacturing practices for yeast extract production; autolysis, plasmolysis and hydrolysis. In this lab. We will discuss the autolysis process and hydrolysis

III.1 Autolysis: Is a process by which the cell components within the cells are solubilized by activation of enzymes that are inherently present in the cell. This is achieved by carefully controlling temperature, pH and time, with the careful addition of enzymes or reagents to stimulate degradation and release of the cell contents into the medium.

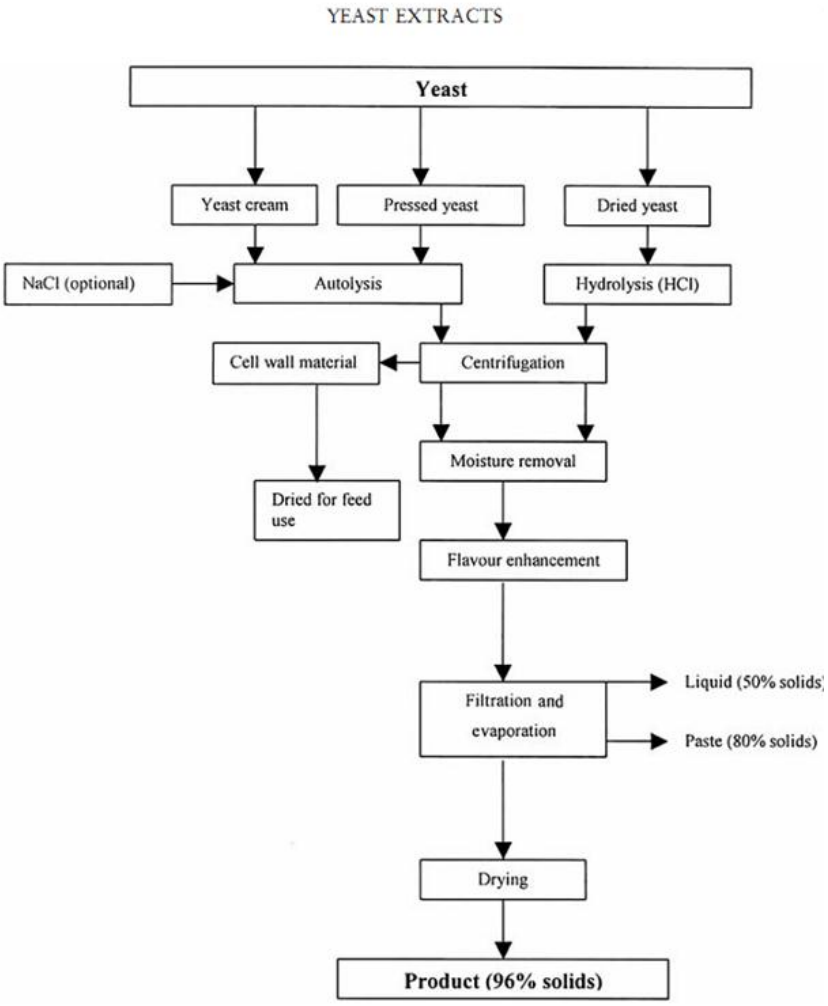


Fig. 7.1 Yeast extract manufacturing process

Procedure:

1. Fresh baker's yeast was suspended in 10 ml distilled water to achieve a concentration of 50%(w/v).
2. The yeast suspensions were placed in 100ml glass vessels immersed in a temperature-controlled water bath with shaker. The initial pH was adjusted to 5.2 with NaOH (10 M).
3. The autolysis was carried out at 52 °C for 72 h. The reaction mixture was shaken at a speed of 120 rpm.

To determine the solid, protein and carbohydrate contents of the autolysate, the reaction mixture was heated in a water bath at 100 °C for 10 min to deactivate the enzyme and centrifuged (3500 rpm) for10 min.

Other procedure:

1. Take 50 g from compressed baker's yeasts then add to them 3 g of sodium chloride to make yeast mud.
2. Then water was added until the total volume became 60 ml. The resultant was used as a control. On the other hand, a series of mud-like materials having the definite ethanol concentrations was prepared by using mixture of ethanol and water instead of water in the control. Those mud-like materials were allowed to stand at 40°C for 72 hours to autolysis. In that case, the sodium chloride concentration of each mud-like material was 5% (w/v).
3. Stirring the solution in order to make homogeneous the whole reaction system comprising yeasts, sodium chloride, ethanol and water.
4. Add water to each mud-like material until the total volume become 100 ml.
5. The resultants were subjected to steam-heating treatment at 100°C for 30 minutes in an autoclave to make enzymatic activities inactive.
6. After cooling to a room temperature, insoluble residues were removed by means of centrifugation to obtain the liquid portion. That is, the autolysate liquid was collected.
7. The residues were suspended in 50 ml of hot water and the washing was collected by means of centrifugation. This procedure was repeated twice.
8. The respective washings were combined with the autolysate liquid and the combined liquid was subjected to concentration under reduced pressure by using a rotary evaporator until the total volume of each combined liquid become 100 ml.

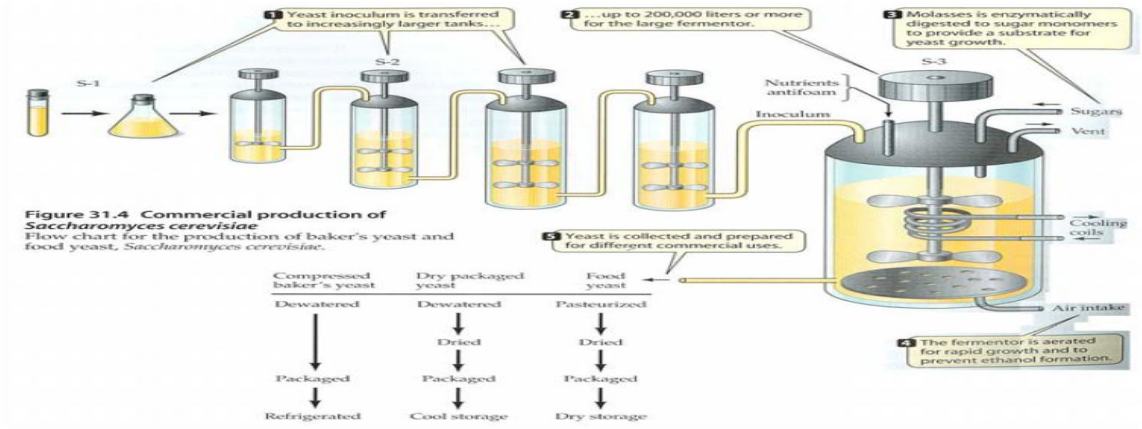


Figure 31.4 Commercial production of *Saccharomyces cerevisiae*
Flow chart for the production of baker's yeast and food yeast, *Saccharomyces cerevisiae*.

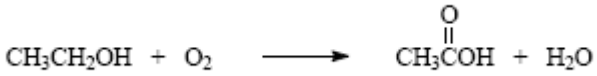
Production of Vinegar

Objectives

- Be able to produce of vinegar by using fermentation techniques.

Introduction

Acetic acid is the chemical compound responsible for the characteristic odor and sour taste of vinegar. Typically, vinegar is about 4 to 8% acetic acid. As the defining ingredient of vinegar, acetic acid has been produced and used by humans since before the dawn of recorded history. In fact, its name comes from the Latin for vinegar, *acetum*. Vinegar is formed from dilute solutions of alcohol, by the action of certain bacteria in the presence of oxygen. Acetic acid bacteria are bacteria that derive their energy from the oxidation of ethanol to acetic acid during respiration. They are Gram-negative, aerobic, rod-shaped bacteria.

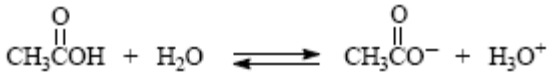


The acetic acid bacteria are found in nature where ethanol is being formed as a result of yeast fermentation of sugars and plant carbohydrates. They can be isolated from the nectar of flowers and from damaged fruit. Other good sources are fresh apple. In these liquids the acetic acid bacteria grow as a surface film due to their aerobic nature and active motility.

Some genera, such as Acetobacter, can eventually oxidize acetic acid to carbon dioxide and water using Krebs cycle enzymes. Other genera, such as Gluconobacter, don't further oxidize acetic acid, as they do not have a full set of Krebs cycle enzymes.

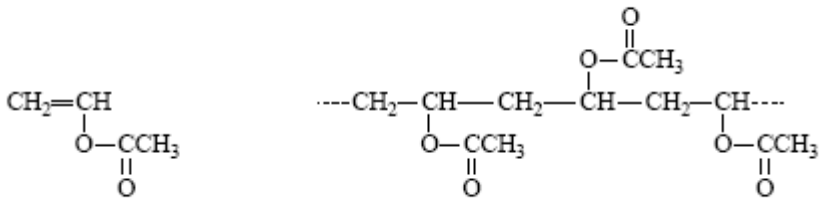
Because it is acidic, vinegar also inhibits the growth of bacteria, so vinegar is used as a preservative in foods, such as pickled vegetables, and as a mild disinfectant in cleaning. Of course, its sour taste, which is also a result of its acidity, makes it popular as a flavoring in cooking and in salad dressings.

Acetic acid is classified as a weak acid, because it does not completely dissociate into its component ions when dissolved in aqueous solution. At a concentration of 0.1 M, only about 1% of the molecules are ionized. In solution, there is a dynamic equilibrium between the neutral molecules and the acetate and hydronium ions.



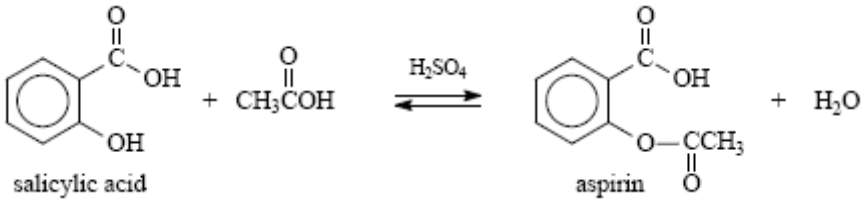
The acid dissociation constant (Ka value) for acetic acid is 1.8 × 10–5 at 25°C.

Acetic acid is an important industrial chemical. The primary use of this chemical is in the manufacture of assorted acetate esters. These are substances formed by reacting acetic acid with a substance containing a hydroxyl (–OH) group. Cellulose, found in cotton and wood, is a polymeric material containing multiple hydroxyl groups. It reacts with acetic acid to yield cellulose acetate, which is used to make films and textiles. Some photographic films are made of cellulose acetate, and rayon is made from cellulose acetate fibers. Vinyl acetate, another ester of acetic acid, polymerizes to form poly(vinyl acetate), which is used in water-based latex paints and in glues for paper and wood.



Acetic acid is also used as a fungicide and as a solvent for many organic compounds.

Acetic acid is also used in the preparation of pharmaceuticals. Aspirin (acetylsalicylic acid) is formed by the



reaction between acetic acid and salicylic acid.

Materials and methods:

Fermentation medium

The medium employed in this lab was previously produced in last week (Production of ethyl alcohol), so we take the fermented product of sugar to make vinegar as following these steps:

I.a For isolation of Acetic acid bacteria (*Acetobacter*) we need:

Apples
Water
Knife
Flask
Goes

Procedure:

1. Wash the apples and cut it into small pieces then put it into a flask.

2. Cover the opening part of flask by goes.

3. Leave the flask for seven weeks for production of vinegar by traditional methods but in our protocol we leave it for one week then isolate the acetic acid bacteria from the surface of the liquid.

I.a.1 For fermentation of acetic acid bacteria:

1. Acetic acid fermentations were carried out in 250 ml Erlenmeyer flasks containing 33 ml of the alcoholic product, obtained from the fermentation of the sucrose supplemented with glucose and 17 ml of bacterial inoculum.

2. Yeast used for the alcoholic fermentation were removed by filtration (0.45 µm pore size) before inoculation with the acetic acid bacteria.

3. Acetic fermentations were carried out at 30°C and 250 rpm

Industrial Production of Acetic Acid

1- Trickling filter

• vinegar manufacturing industry near Orleans in 14th century
• trickling filter, wooden bioreactor (volume up to 60 m3) filled with beechwood shavings, acetic acid bacteria grow as biofilm
• the ethanolic solution is sprayed over the surface and trickles through the shavings into a collection basin, and recirculated
• temperature maintained at 29-35°C
• about 12% acetic acid produced in 3 days

• the life of a well-packed and maintained generator is about 20 years

2- Submerged, batch process (Frings acetator)

• stainless steel tank with a high-speed mixer
microbes, air, ethanol and nutrients are mixed to provide a favorable environment for microbial growth
• 30°C maintained by circulation of cooling water
• 12% acetic acid in about 35 h
• production rate per m3 over 10 times higher than with surface “fermentation” and over 5% higher than with trickling filter

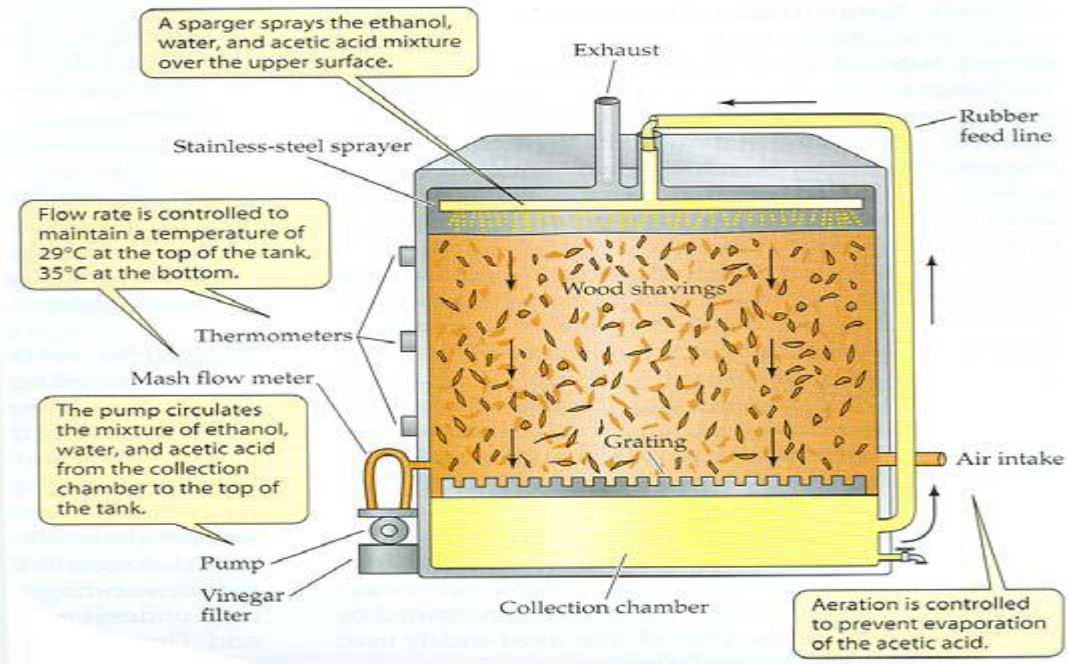


Figure 31.5 Manufacture of vinegar by continuous fermentation
Frings-type trickling generator for the commercial production of vinegar.

Production of Lactic acid

Objectives

- Be able to produce lactic acid from cheese whey by using fermentation techniques.
- Be able to detect lactic acid by chemical test.

I. Introduction

Cheese whey is the liquid effluent generated during the cheese making process. However, cheese whey contains about 5% lactose, which is a suitable substrate for the production of value added products using biochemical conversion processes.

Lactic acid is a three carbon organic acid and exists in two optically active isomeric forms with the structure shown in Fig. 10.1

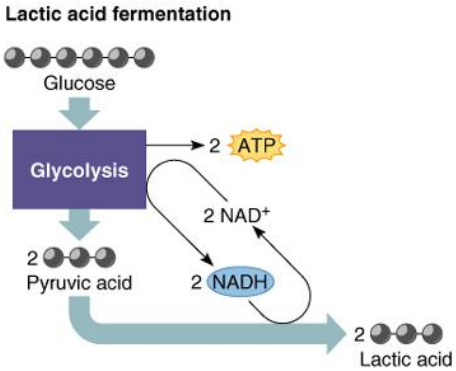


Fig. Optical Isomers of Lactic Acid.

Lactic acid is soluble in water and water miscible organic solvents but insoluble in other organic solvents.

Lactic acid is a commercially viable product. It is used in such things as: meat and poultry preservation, cosmetics, oral and health care products and baked goods. One way to produced lactic acid is through the fermentation of sugar from the microorganism *Lactobacillus*. Under optimal conditions of 37ºC and with the sugar glucose or lactose, *Lactobacillus* will convert these sugars into lactic acid.

Fig. Lactic Acid fermentation.



Lactic acid fermentation

• pyruvate is reduced to lactic acid with the coupled
• lactic acid bacteria (e.g. *Lactobacillus*, *Streptococ* fermentations
• fermented milk, cheese, fermented vegetables

Homolactic fermentation

• glucose degraded via EMP pathway, with lactic acid as the only end product

glucose + 2 ADP + 2 P_i ➡ 2 lactic acid + 2 ATP

• carried out by *Streptococcus*, *Pediococcus*, *Lactococcus*, *Enterococcus* and various *Lactobacillus* species
• important in dairy industry (yogurt, cheese)

Heterolactic fermentation

• glucose degraded via pentose phosphate pathway
• in addition to lactic acid, also ethanol and CO2 produced

glucose + ADP + P_i ➡ lactic acid + ethanol + CO_2 + ATP

II. Production of lactic acid by fermentation technique:

Materials and methods:

Cheese whey	450 ml
Bottles	3
Glucose	1%
Lactose	1%
Glucose + Lactose	1% +1%

1. Lactic acid fermentations were carried out in 350 ml bottle containing 150 ml of the cheese whey.

2. The first one will contain 150 ml cheese whey only.

3. Add 1% of glucose to the second one, 1% lactose to the third bottle and 1% of glucose + 1% of lactose to the fourth one.

4. Detect the pH for every bottles of fermentation.

5. Autoclave at 121°c, 1.5 atm for 15- 20 min.

6. When solutions have cooled to below 50ºC, add 2 ml Yoghurt solution for (*Lactobacillus*).

7. Incubate the bottles for 3- 4 days at 37ºC and 200 rpm.

8. Detect the pH for every day and compare it with the first detection.

9. Record your results in the table below and make a graph for these results using excel program.

	Date and Time	Hours	pH
1% Glucose			

	Date and Time	Hours	pH
1% Lactose			

	Date and Time	Hours	pH
1% Glucose + %Lactose			

I.a Detection of lactic acid by chemical test:

Materials and methods:

1. Take 1.5 ml from the first fermented bottle.

2. Add drop of NaOH 1% for making the solution alkaline (use litmus paper).

3. Add 5 ml of 10% KI.

4. Finally add 5 ml of NaClO.

Results:

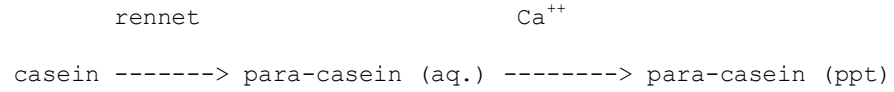
Production of Cheese

Objectives

To demonstrate the use of rennet in casein coagulation in different pH conditions.

Introduction

Cow's milk is rich in a wide range of chemical compounds that can be processed into various dairy products such as cheese, butter, and yogurt. Specifically the milk component involved in cheese production is a soluble protein called casein. The enzyme rennet can be used to catalyze the conversion of casein in milk to para-casein by removing a glycopeptide from the soluble casein. Para-casein further clots, i.e. coagulates, in the presence of calcium ions to form white, creamy lumps called the *curd*, leaving behind the supernatant called the *whey*.



The precipitate is soft at this point and can be separated from the whey by the use of cheese cloth. Filtration does not work very well; filter paper clogging is a recurrent problem.

There is no standard method of cheese making; limitless variations exist for all stages of the process: pre-ripening, curdling, addition of artificial ingredients and salt for flavor, and aging. This variation in processing accounts for the wide range of cheeses commercially available, differing in texture and flavor. The curd can also be processed with other techniques to make a variety of desserts. However, all processes have one thing in common: the separation of the curd from the whey.

Three ways of preparing milk for curdling will be introduced in this experiment. Industrially, the lactic acid level in the milk is increased by adding a starter culture of *Streptococci* or *Lactobacilli* to the milk and fermenting at 32ºC for 10 to 75 minutes. In addition to biologically converting the lactose present in the milk to lactic acid, these strains of microorganisms also greatly affect the flavor of the final product. Thus, the selection of a suitable strain, the amount of starter culture, and the length of pre-ripening, is of the utmost importance in creating the subtle differences in the final color and aroma that distinguishes an expensive cheese from a cheap one.

If one has not yet acquired a keen palate for cheeses, the second approach should suffice. In this approach, one takes advantage of the existing *Lactobacillus* culture in buttermilk and uses it as the starting culture. One ml of buttermilk is added per 100 ml of milk, and the mixture is then fermented at room temperature for 4-12 hours. At the end of fermentation, the temperature of the mixture is raised to 32ºC, and artificial coloring is added to the mixture prior to curdling.

The third way to prepare the milk in a short time frame is to add acid (HCl) and to heat to 32ºC. Of course, there leaves much to be desired in this method if you are a cheese connoisseur.

After rennet is added to the pre-cured milk, the coagulation process is started. In cheese making, as coagulation comes to completion, the temperature is gradually raised to about 38ºC. This slightly elevated temperature facilitates the separation of the curd from the whey. A higher temperature also hardens the curd. The curd may be hardened further by cooking it for a longer period of time, either with or without the whey.

After the curd is separated from the whey, salt, seasoning, and other curing and flavoring ingredients are added. The curd is wrapped in cheese cloth and pressed for 12 to 18 hours to remove the additional whey soaked in the curd. The curd hardens and forms a cheese block in the shape of the press as the whey is squeezed out. Finally, the cheese block is dried for 6 hours.

Precuring at a Warm Temperature (Acidify the Milk)

⇓

Coloring

⇓

Coagulation by Rennet

⇓

Separation of the Whey

⇓

Flavor addition

⇓

Compression of the Curd

⇓

Aging

⇓

Finished Cheese

Figure 1. Steps in cheese making.

It is now ready for consumption, or it may be left to age in a controlled cool environment (2-13ºC). Although a higher temperature promotes faster curing, there is also a higher chance of spoilage due to undesirable microbial activities at elevated temperatures. Prior to aging, the cheese block is usually wrapped tightly to exclude air and microbial contaminants from entering and spoiling the cheese. One way to accomplish this is to dip the cheese block in a pot of melted wax. During the aging process, many complicated microbial and chemical actions continue to take place in the cheese block. Thousands of techniques exist to develop various distinctive flavors. These reactions are not well characterized; thus, cheese making is still an art rather than a science. Depending on the technique employed, this final aging process takes any where from 2 weeks to 6 months.

Cheese making

Pre-ripening

• inoculation with starter culture of lactic acid bacteria
• acidification by fermentation of lactose to lactic acid

Coagulation of milk proteins (casein)

• addition of chymosin (rennin), an acid protease from calves' stomach - *renneting*
since 1980s also recombinant enzyme produced in yeast
• catalyzes specific hydrolysis of κ-casein leads to coagulation in the presence of Ca^{2+} to form a gel (curd)
• also traps bacteria that continue lactic acid fermentation
• removal of liquid whey from the curd
• addition of secondary microbiota
• treatments vary for different cheese varieties

Lactic acid has the following effects:

- It causes the coagulation of casein at pH 4.6, the isoelectric point of that protein, which is used in the manufacture of some cheeses, e.g. cottage cheese.
- It provides a favorably low pH for the action of rennin the enzyme which forms the curd from casein in other types of cheeses.
- The low pH eliminates proteolytic and other undesirable bacteria.
- It causes the curd to shrink and thus promotes the drainage of whey.
- Metabolic products from the lactic acid bacteria such as ketones, esters and aldehydes contribute to the flavor of the cheese.

List of Reagents and Instruments

A. Equipment

- 2 beakers, 400 ml
- Graduated cylinder
- Pipet or medicine dropper
- Temperature bath or heat source
- Thermometer
- Stirring rod
- Cheese cloth
- Balance

B. Reagents

- Pasteurized milk, (to be supplied by the student)
- Butter milk or lactic acid producing strains, (to be supplied by the student)
- Rennet enzyme

Procedures

1. The T.A. will prepare several batches of cultured milk before the class by mixing 2.5ml of buttermilk (starter culture) to 250ml of normal pasteurized milk in a beaker. The content in the first beaker is to be fermented at room temperature for 12 hours (G); the second for 10 hours (F); the third for 8 hours (E); etc. Do not add buttermilk to the batch labelled A, which is kept free of a starter culture. The number of mixtures with differing pre-curing times depends on the number of students enrolled. Each student will be responsible for analyzing one of these beakers.
2. Analogous to Step 1., prepare several synthetic "cultured" milk by adding 5-20 ml of 1N HCl to 250 ml of milk in a beaker. Each student will be responsible for one of these beakers. (What happens when lactic acid or citric acid is used in lieu of HCl?)
3. Record the pH of each batch.
4. Gently heat the authentically and synthetically cultured milk to 32ºC in a temperature bath.
5. Stir and add 0.5 ml (approx. 5 drops) of rennet to each beaker.
6. After the enzyme addition, let the milk stand on the bench top for 30 minutes to allow coagulation to proceed. Do not disturb the coagulation process in the beakers; otherwise, the curds will be broken into pieces too small to filter efficiently with cheese cloth.
7. After curd formation, break up the curds with a stirring rod and drain the whey through layers of cheese cloth. Filtration through a medium pored filter paper may give better results.
8. Weigh the amount of curd obtained in its wet state and approximate the yield.
9. Taste your cheese. (Yum-m-my, I love cheese!) Is there anything missing?

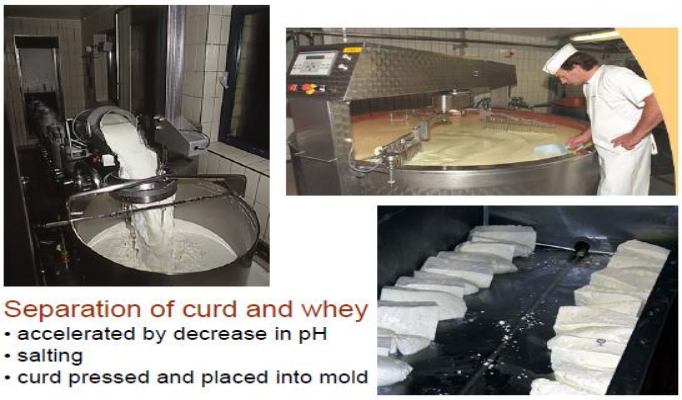
Discussions

It is generally quite straight forward to locate the sources of enzymes; the natural sources of enzymes are where nature needs them. It comes as no surprise that rennet had traditionally been isolated from the fourth stomach of young calves because digestion by suckling calves is nature's primary way of processing cow milk. However, the ways of nature are not always the most economical from man's perspective when a process is adapted to different uses to benefit mankind in ways not originally intended by the nature. Thus enters the study of biochemical engineering.

Chemical engineering is the study of how to make a large quantity of chemicals in an economical fashion. For example, frequently a chemical engineer must devise a process to mass-produce a polymer that is totally different from the one originally used by an organic chemist working with small test tubes and beakers in a laboratory. Biochemical engineering, being a sub-field of chemical engineering, also deals with the same kinds of problems facing chemical engineering, except that the chemicals are not synthetic (manmade) but biological (naturally existing) in nature. Although rennet is naturally excreted from a calf's stomach lining, extracting it from its natural source is not economical. Other proteases can also convert casein to para-casein, but their action does not stop there. They further degrade the curd to soluble subunits. Fortunately, large quantities of rennet of consistent quality can now be produced easier and cheaper in a well controlled environment by microbial fermentation.

A word of caution is in order here. Enzymes as a class of chemicals are not generally considered as dangerous, toxic, nor poisonous; they do not cause skin irritations or burns as acids or bases. Some exceptions are proteases that catalyze the breakdown of protein molecules to amino acids components. Because meat is mainly protein, protease can digest the soft moist sections of the skin. We can all imagine how that is going to feel. Thus, you should exercise the same caution with enzymes as you do with any other chemicals.





Production of Yoghurt

Objectives

- Be able to produce of yoghurt from cheese whey using *Lactobacillus delbrueckii* and *Streptococcus thermophilus*.

I. Introduction

Yoghurt or **yogurt**, less commonly **yoghourt**, **joghurt** or **yogourt**, is a dairy product produced by the controlled fermentation of milk by two species of bacteria (*Lactobacillus* sp. and *Streptococcus* sp.). The sugar in milk (called lactose) is fermented to (lactic acid) and it is this that causes the characteristic curd to form.

I.1 Yogurt Types:

1. **Set yogurt**: A solid set where the yogurt firms in a container and not disturbed.

2. **Stirred yogurt:** Yogurt made in a large container then spooned or otherwise dispensed into secondary serving containers. The consistency of the "set" is broken and the texture is less firm than set yogurt. This is the most popular form of commercial yogurt.

3. **Drinking yogurt:** Stirred yogurt to which additional milk and flavors are mixed in. Add fruit or fruit syrups to taste. Mix in milk to achieve the desired thickness. The shelf life of this product is 4-10 days, since the pH is raised by fresh milk addition. Some whey separation will occur and is natural. Commercial products recommend a thorough shaking before consumption.

4. **Fruit yogurt:** Fruit, fruit syrups, or pie filling can be added to the yogurt. They are placed on top, on bottom, or stirred into the yogurt.

I.2 Health benefit of yogurt

1. Yogurt is easier to digest than milk.
2. Yogurt contributes to colon health.
3. Yogurt remains an excellent source of calcium.
4. Yogurt can boost immunity.
5. Yogurt aids healing after intestinal infections.
6. Yogurt can lower LDL ,Raise HDL cholesterol.

II. Method of processing:

II.1 Yoghurt

There are two types of plain (unflavored) yoghurt that are produced under control conditions. These are set that yoghurt that has a firm crud and stirred yoghurt that has a smooth creamy consistency, similar to a thick cream. The production method is:

1. Heat milk in a stainless steel (or less desirably an aluminum) pan to 80- 85 ºC for 15- 20 min. with constant stirring.

2. Cool 40- 45 ºC as quickly as possible and add a starter culture (or inoculum) of the mixed lactic acid bacteria.

3. Depending on the yoghurt required, the milk is either kept in pan (for stirred yoghurt) or filled into individual retail pots (for set yoghurt).

4. Incubate the milk at 40- 45 ºC for 4- 6 hrs to allow the fermentation to take place.

5. For stirring yoghurt, the pan of the crude is then stirring using a sterilized spoon or ladle until the crude becomes a smooth cream that has a thick consistency without any lumps.

6. Both types of yoghurt are stored in a refrigerator at 4- 8 ºC.

Flow chart for commercial production of stirred yoghurt

Heat milk to 85-95°C for 15-30 minutes

↓

Homogenise the milk

↓

Cool the milk to incubation temperature

↓

Add the bacterial starter

↓

Incubate the mixture at 40-46°C for 4-5 hours

(until the pH has fallen to around 4.3)

↓

Stir the thickened yoghurt and cool it to 5°C

↓

Add fruit, flavourings, colorants (if required)

↓

Pack and despatch the yoghurt, keeping it refrigerated

II.2 Flavoured or fruit yoghurt

Flavoured or fruit yoghurt is produced the same as plain yoghurt. For flavoured yoghurt, the essence (e.g, vanilla, chocolate or fruit flavour) with or without coloring, is added befor the milk is fermented.

Yoghurt that contains fruit pulp is made in the similar way to plain yoghurt and the fruit pulp is added after fermentation. Less acidic fruits such as melons or pawpaw (papaya) are more successful because they don't react with the milk. Acidic fruits (lemons, lime, pineapple, etc.) may cause excessive curdling and separation of the yoghurt.

II.2.1 To make fruit pulp

1. Select fully rip fruit.
2. Wash in clean water.
3. Remove the peel and seed.
4. Pulp the fruit.
5. Heat the pulp for 15- 20 min. at 70- 80 ºC in stainless steel pan, with constant stirring to prevent it burning in to the pan. This softens the fruit and destroys any contaminating microorganisms and naturally occurring enzymes that would cause changes to the color or flavour of the fruit during the storage.
6. Cool to room temperature.
7. Either mix the pulp into stirred yoghurt (e.g. one part pulp between three to five parts yoghurt) or place a layer of pulp at the bottom of pots before filling stirred yoghurt.
8. Store as for plain yoghurt.

Production of Ethyl Alcohol

Objectives

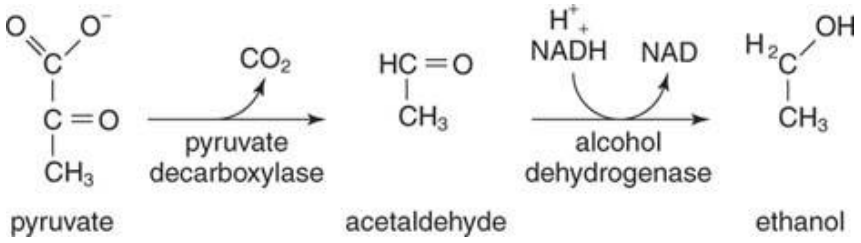
- Be able to produce of ethyl alcohol by using fermentation techniques.
- Be able to detect of ethyl alcohol by chemical test.
- Be able to distillate the production by simple distillation method.

Introduction

Ethanol fermented from renewable sources for fuel or fuel additives are known as bio-ethanol. Additionally, the ethanol from biomass-based waste materials is considered as bio-ethanol.

Both yeasts and bacteria have been used for the production of ethanol. Among the bacteria, the most widely used organism is *Zymomonas mobilis*. *Saccharomysis cerevisiae* is the most commonly used yeast but *Kluyveromyces fragilis* has also been employed.

Under aerobic conditions and the presence of high glucose concentration, *Saccharomysis cerevisiae* grow well but produces no alcohol. Under anaerobic condition, however, growth slows and pyruvate from glycolytic pathway is split with pyruvate decarboxylase into acetaldehyde and Co_2. Ethanol is then produced from acetaldehyde by reduction with alcohol dehydrogenase.



Ethanol is produces in three steps, each of which must be optimized: 1- Preparation of nutrient solution; 2- fermentation; 3- distillation of ethanol.

I. Preparation of the nutrient solution:

Materials:

I.a For 1L of Sucrose nutrient broth you need:

Sucrose	20%
$MgSo_4$	0.5g
KH_2Po_4	1g
$(NH4)_2So_4$	5g
Yeast extract	1g

Adjust pH to 4.

I.a.1 For 1L of Glucose nutrient broth you need:

Glucose	10g
Peptone	5g
KH_2Po_4	2g
Yeast extract	20g

Procedure:

1- You will prepare from I.a 700 ml and from I.a.1 70 ml only.

2- Autoclave at 121°c, 15 atm for 15- 20 min.

3- For I.a.1 add 0.5g of baker's yeast aseptically after cooling then Incubate the media in water bath at 37°c for 1- 2 hr for activation.

4- Add 70 ml of glucose nutrient broth to 700 ml of sucrose nutrient broth aseptically.

5- Finally incubate the whole media at 37°c for one week.

II. Detection of ethyl alcohol by chemical test:

Materials and procedures:

1- Add 5 ml of $K_2Cr_2O_7$ to 1 ml of H_2SO_4 then heat slowly.

2- Add 1 ml from the fermented media.

Results:

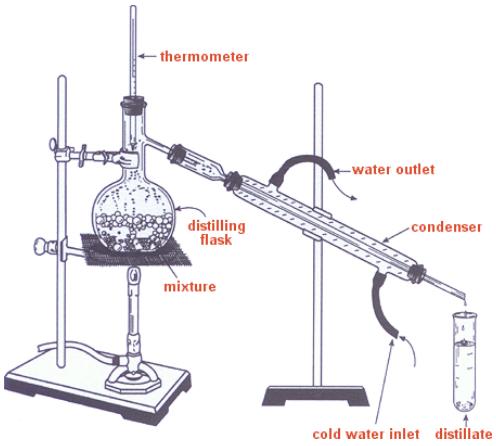
III. Distillation the production:

A method of separating two or more liquids, by changing the equilibrium balance between them. Ethanol boils at 78.5° C, so it's pretty easy to get the water out.

Heat the alcohol/water mixture, then when the ethanol evaporates, it's caught and condensed, producing a higher concentration of alcohol. It can be run through the still several times, each time producing a higher concentration.

Procedure:

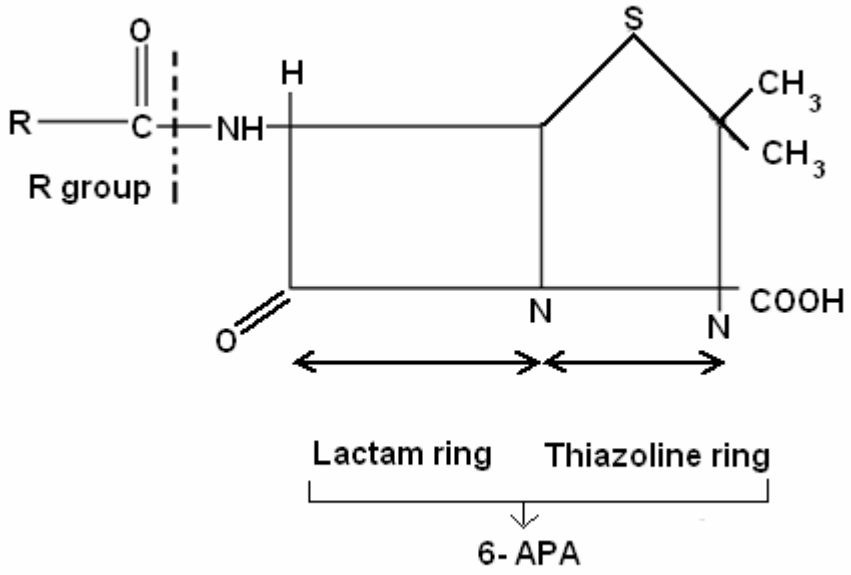
Set up a distillation apparatus as in **Fig. 8.1**



Commercial Production of Penicillin *(Sub Merged Fermentation)*

Penicillin is one of the first antibiotics to be discovered. There are many types of penicillins, but the important ones are penicillin V and penicillin G. Some of the penicillins are naturally

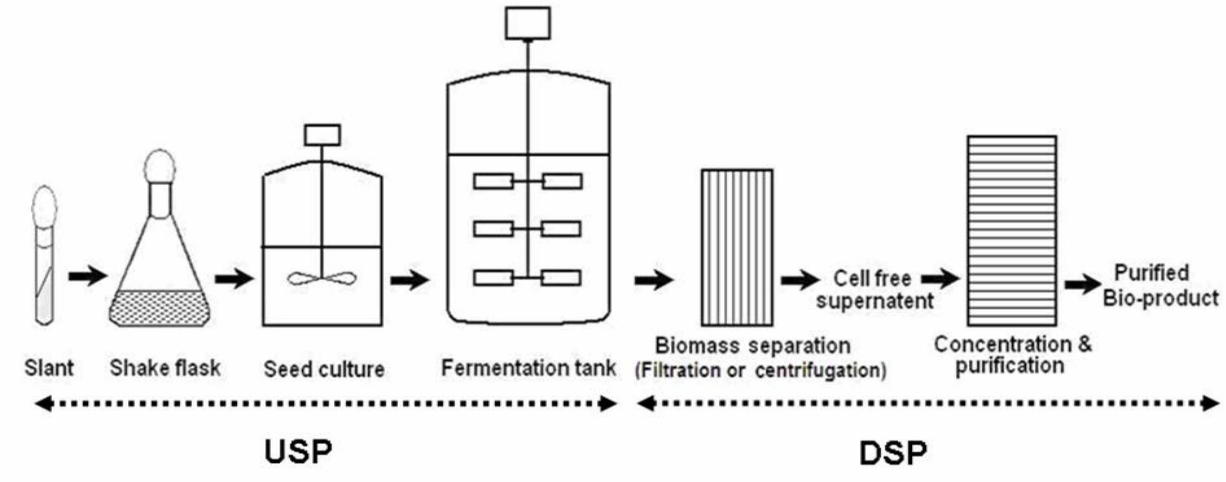
occurring and others are man made. Naturally occurring penicillins have a basic structure composed of 6-aminopenicillanic acid (6-APA) which is made up of β-lactam ring fused to thiazolidine ring (Fig. 17). The 6 amino position of β-lactam ring carries a variety of acyl moieties.



6-APA: 6-Aminopenicillanic acid

Fig. 17: Structure of β–lactam antibiotic, penicillin

Penicillins are bactericidal in nature and are highly effective against Gram positive bacteria like *Staphylococcus, Neisseria* etc. The major producer of penicillins are fungi especially *Penicillium chrysogenum*. Penicillin exhibits the property of a typical secondary metabolite being produced at the end of log phase. The yield of penicillin has been increased tremendously by the use of classical mutations, genetic engineering and media optimization studies. At industrial level yields of 50g/litre of growth medium have been achieved



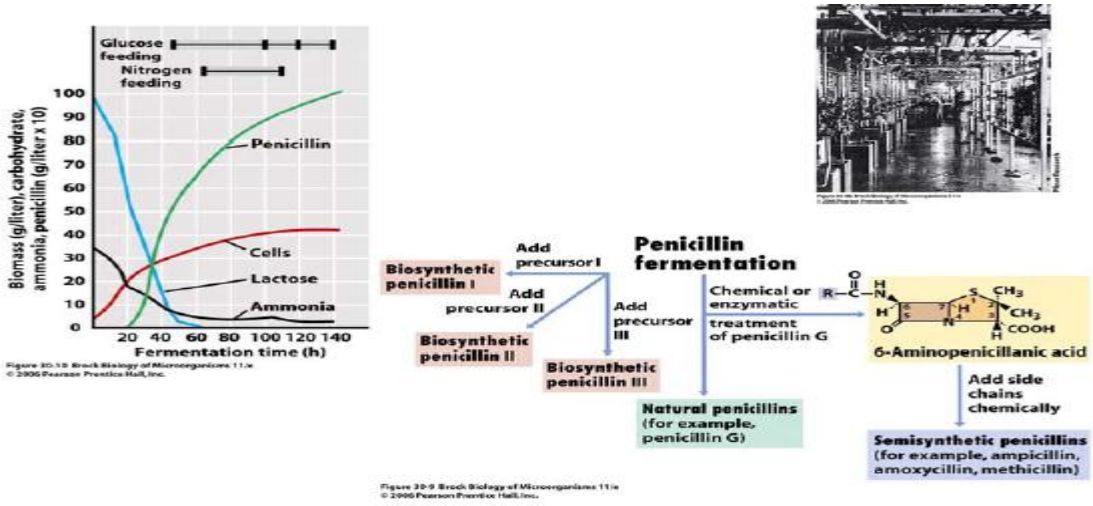
USP: Up stream processing; DSP: Down stream processing

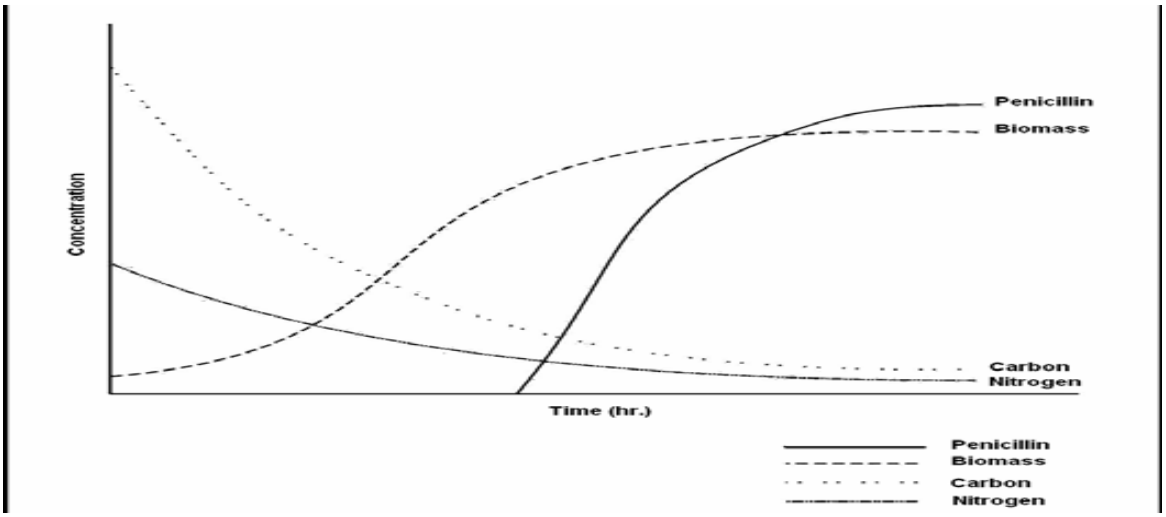
Fig. 18: Upstream and downstream processing in penicillin fermentation

Commercial production of penicillin

Fig. 18 shows a birds eye view of the production, purification and recovery of penicillin. In general, commercial penicillin is produced by fed-batch fermentation process using bioreactors of 30K-250K capacity. The fermentation temperature is maintained around 25-27$_o$C, pH 6.5-7.0 and a constant supply of oxygen (25-60mmoles/l/hr) depending upon the type of *P.chrysogenum* used. The fermentation medium is composed of a carbon source like lactose and a nitrogen source like corn steep liquor. Additional supplements like yeast extract, soy meal, ammonium salts are also added. In addition, side chain precursors like phenyl acetic acid or phenoxyacetic acid are also added to the medium. The fermentation

process involves a vegetative growth phase (around 40 hrs) followed by penicillin production phase (150-180 hrs, Fig. 19).





The culture inoculum is prepared by transferring spore suspension of *P.chrysogenum* into a 100-200ml liquid medium. The vegetative inoculum thus prepared is transferred to 1-2 litres

fermenters. The mycelium is then sub-cultured for one or two stages with increasing the volume of fermentation till sufficient inoculum is ready for penicillin production. Care is taken to prepare the fungal inoculum in loose form rather compact pellet. The latter condition of mycelium is indicative of metabolically poor fungal biomass.

Recovery and purification of penicillin

The fermentation broth is passed through filters to remove fungal mycelium. The pH of the broth is adjusted around 3.5 and is mixed with organic solvent (amyl acetate or butyl acetate or methyl ketone) which extracts the penicillin from aqueous phase under cold conditions. The penicillin rich solvent is passed through activated charcoal to remove impurities and pigments. Subsequent addition of sodium or potassium acetate results in crystallization of penicillin as sodium or potassium penicillin. The crystals are then recovered by rotary vacuum filtration. By this procedure penicillin is more than 90% pure.

Enzyme / Cell Immobiliztion

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In the course of the last few years, enzymes have gradually gained preference over their chemical counterparts, especially in the food and pharmaceutical industries. This may be attributed to the high stereo specificity of enzymes which in turn contributes to a decrease in the proportion of undesirable side products. Moreover, the existing biodiversity more than ensures the possibility of obtaining an enzyme for a particular reaction. However, there exist significant challenges in the use of enzymes on a commercial basis. Very often, the conditions employed in the bioprocess, lead to the denaturation of the enzyme. Moreover, the high cost of enzymes can render a process financially nonviable. This is where the concept of immobilization has stepped in and provided a means to combat some of the challenges faced by the industry in the use of enzymes on a commercial scale.

Immobilization essentially involves, the stable association of an enzyme with an insoluble and inert support, so as to render the enzyme stable and facilitate its reuse. Moreover, the product obtained is not contaminated with the enzyme, thus leading to a significant reduction in the costs involved in the downstream processing. In case of immobilization, there exists the possibility of using either isolated enzymes or even whole cells. The latter affords the convenience of doing away with the lengthy procedures involved in the purification of enzymes and also ensures the functionality of the enzyme since its natural environment remains unaltered. However, unwanted side reactions may occur, in which case, immobilization of the pure enzyme may be preferred.

Support material

The choice of the support matrix is a very important consideration, the parameters for which are defined, based on the chemical nature of the enzyme, the substrate, the type of reaction, as well as the intended use of the product. Some of the criteria used in the selection of an appropriate support include physical parameters like, the strength of the matrix, the available surface area, the degree of porosity, and the flow rate as well as the chemical properties such as its hydrophillicity, inertness and the available functional groups. Moreover, there are other important considerations which include, the susceptibility of the support to microbial attacks

or disruption due to the chemicals, pH, temperature or the action of proteases, as well as its availability and cost, thus determining the feasibility of its usage during scale up. A very important point to be taken care of is that, the matrix should not render the product toxic, especially if the product is meant for consumption.

Principles used in immobilization

Adsorption: This is one of the simplest means of immobilization, and involves electrostatic forces such as vanderwaal forces, ionic bridges and hydrogen bonds, which are instrumental

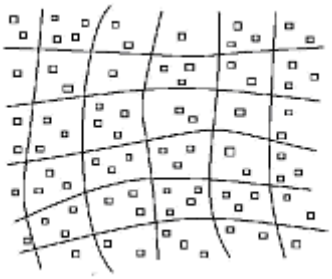
in establishing the reversible surface interactions between the support and the enzyme/cells (Fig. 20). Hydrophobic interactions, though weak, are enough in number to ensure sufficiently reasonable binding. The procedure basically involves mixing of the biological component with the matrix for a fixed period of time at an optimum pH, followed by washing, in order to get rid of the non biological components. Here, the catalytic properties of the enzyme remain unaltered, since no change is brought about in the chemical nature of either the enzyme or the supporting matrix. Moreover it serves as a rapid and economical means of immobilization, allowing the regeneration of matrix, since it is a reversible process. However, a major limitation of this method lies in the possibility of the enzyme leaking from the support owing to changes in the reaction conditions.



, enzyme; - - - -, electrostatic forces; matrix.

Fig. 20: Enzyme immobilization by Adsorption technique

Entrapment: Herein, the enzyme molecules are free in solution but their movement is restricted owing to the lattices in the gel (Fig. 21). The degree of porosity is set so as to prevent leakiness of the enzyme but allows unrestricted diffusion of the substrates as well as the products.



, enzyme molecules; , matrix.

Fig. 21: Enzyme immobilization by Entrapment technique

Entrapment may be achieved by a number of means which include the following:
i) Ionotropic gelation of macromolecules with multivalent cations: This involves mixing of the polyionic polymer with the enzyme, followed by cross linking of the polymer with multivalent cations, in an ion exchange reaction, to form the required lattice structure.
ii) Temperature induced gelation: This is brought about by adding the enzyme to a 1-2 % preparation of agarose or gelatin kept at 45°C, and then bringing down the temperature.
iii) Organic polymerization by chemical/photochemical reaction: Here, the enzyme is mixed with chemical monomers which are polymerized to form a cross linked network, trapping the enzyme.

Applications of immobilization technique

Immobilization of enzymes/cells has found varied applications in industry. Some of the most important applications of these in the food industry have been the conversion of glucose to high concentration fructose syrups by the enzyme glucose isomerase, the immobilized invertase systems and the production of lactose hydrolyzed milk by use of β-galactosidase. In the case of the pharmaceutical industry, the production of 6-amino penicillanic acid or 7-amino deacetoxy cephalosporanic acid by means of immobilized systems is seen as a major achievement. Immobilized proteases offer a great deal of applications in the field of synthesis and modification of peptides as well as in the leather industry. Moreover, they have been used as biosensors. Immobilized glucose oxidase and peroxidase enzymes have been used as detection systems for serum glucose levels. Uricase has been used in a similar fashion for detection of uric acid levels while cholesterol esterase and cholesterol oxidase have been used for detection of serum cholesterol. However, while a number of exciting possibilities do exist in this field, for an immobilized system to succeed on an industrial scale, there are a number of considerations such as, the costs involved, the feasibility of using a particular matrix, as well as the purity of the end product